

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399958
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Patel; Paresh K.

Systems and methods for tool activation and display cabinet locking

Abstract

Disclosed are systems and methods for tool activation and display cabinet locking. In one aspect, a method for activating a product is performed at the product. The product includes a power supply, a controller, a persistent memory coupled to the controller, and a radio module. The product is configured to be operable in a plurality of states, and a state of the product is in an inactivated state. The method includes receiving via the radio module an activation message and an authorization token. The method includes attempting to validate the authorization token based on a preset authorization information and the state of the product. The method includes, (i) when the validation is not successful, rejecting the activation message without changing the state of the product; and (ii) when the validation is successful, changing the state of the product from the inactivated state to an activated state.

Inventors:	Patel; Paresh K. (Portland, OR)
Applicant:	PAYRANGE INC. (Portland, OR)
Family ID:	1000008780060
Assignee:	PAYRANGE LLC (Portland, OR)
Appl. No.:	19/019281
Filed:	January 13, 2025

Prior Publication Data

Document Identifier	Publication Date
US 20250148057 A1	May. 08, 2025

Related U.S. Application Data

continuation parent-doc US 17443802 20210727 US 11935051 child-doc US 18610033
continuation parent-doc US 16934933 20200721 US 11074580 child-doc US 17443802

Publication Classification

Int. Cl.: G06F21/00 (20130101); G06F13/00 (20060101); G06Q20/18 (20120101); G06Q20/32 (20120101); G06Q20/36 (20120101); G06Q20/38 (20120101); G06Q20/40 (20120101); G06Q30/06 (20230101); G07F9/00 (20060101); G07F9/02 (20060101)

U.S. Cl.:

CPC G06F21/00 (20130101); G06F13/00 (20130101); G06Q20/18 (20130101); G06Q20/32 (20130101); G06Q20/322 (20130101); G06Q20/3226 (20130101); G06Q20/327 (20130101); G06Q20/3278 (20130101); G06Q20/36 (20130101); G06Q20/38 (20130101); G06Q20/3821 (20130101); G06Q20/3823 (20130101); G06Q20/3829 (20130101); G06Q20/40 (20130101); G06Q20/401 (20130101); G06Q20/405 (20130101); G06Q30/06 (20130101); G07F9/002 (20200501); G07F9/023 (20130101); G06Q2220/00 (20130101)

Field of Classification Search

CPC: G06F (21/00); G06F (13/00); G06F (9/002); G06F (9/023); G06Q (20/18); G06Q (20/32); G06Q (20/322); G06Q (20/3226); G06Q (20/327); G06Q (20/3278); G06Q (20/36); G06Q (20/38); G06Q (20/3821); G06Q (20/3823); G06Q (20/3829); G06Q (20/40); G06Q (20/401); G06Q (20/405); G06Q (30/06); G06Q (2220/00)

USPC: 705/71

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
309219	12/1883	Fruen	N/A	N/A
3457391	12/1968	Yamamoto	N/A	N/A
3931497	12/1975	Gentile et al.	N/A	N/A
4374557	12/1982	Sugimoto et al.	N/A	N/A
5479602	12/1994	Baecker et al.	N/A	N/A
5734150	12/1997	Brown et al.	N/A	N/A
5844808	12/1997	Konsmo et al.	N/A	N/A
5854994	12/1997	Canada et al.	N/A	N/A
5880733	12/1998	Horvitz et al.	N/A	N/A
5892900	12/1998	Ginter et al.	N/A	N/A
5955718	12/1998	Levasseur	N/A	N/A
6056194	12/1999	Kolls	N/A	N/A
6332127	12/2000	Bandera et al.	N/A	N/A
6390269	12/2001	Billington	N/A	N/A
6462644	12/2001	Howell	N/A	N/A
6505095	12/2002	Kolls	N/A	N/A
6584309	12/2002	Whigham	N/A	N/A

6594759	12/2002	Wang	N/A	N/A
6743095	12/2003	Cole et al.	N/A	N/A
6793134	12/2003	Clark	N/A	N/A
6810234	12/2003	Rasanen	N/A	N/A
6840860	12/2004	Okuniewicz	N/A	N/A
7085556	12/2005	Offer	N/A	N/A
7110954	12/2005	Yung et al.	N/A	N/A
7127236	12/2005	Khan et al.	N/A	N/A
7131575	12/2005	Kolls	N/A	N/A
7455223	12/2007	Wilson	N/A	N/A
7458510	12/2007	Zhou	N/A	N/A
7464867	12/2007	Kolls	N/A	N/A
7493288	12/2008	Bishop et al.	N/A	N/A
7513419	12/2008	Crews et al.	N/A	N/A
7672680	12/2009	Lee et al.	N/A	N/A
7690495	12/2009	Kolls	N/A	N/A
7721958	12/2009	Belfer et al.	N/A	N/A
7848980	12/2009	Carlson	N/A	N/A
7962369	12/2010	Rosenberg	N/A	N/A
7965693	12/2010	Jiang et al.	N/A	N/A
7983670	12/2010	Elliott	N/A	N/A
8020763	12/2010	Kowalchyk	N/A	N/A
8059101	12/2010	Westerman	N/A	N/A
8157167	12/2011	Cost et al.	N/A	N/A
8201736	12/2011	Majer	N/A	N/A
8255323	12/2011	Casey et al.	N/A	N/A
D669899	12/2011	Cheng et al.	N/A	N/A
8346670	12/2012	Hasson et al.	N/A	N/A
8356754	12/2012	Johnson et al.	N/A	N/A
8376227	12/2012	Hammad et al.	N/A	N/A
8396589	12/2012	Katzenstein Garibaldi	N/A	N/A
8412626	12/2012	Hirson et al.	N/A	N/A
8438066	12/2012	Yuen	N/A	N/A
8479190	12/2012	Sueyoshi et al.	N/A	N/A
8489140	12/2012	Weiner et al.	N/A	N/A
8496168	12/2012	Miller et al.	N/A	N/A
8514775	12/2012	Frecassetti et al.	N/A	N/A
8517766	12/2012	Golko et al.	N/A	N/A
8548426	12/2012	Smith	N/A	N/A
8577734	12/2012	Treyz	N/A	N/A
8583496	12/2012	You et al.	N/A	N/A
8596528	12/2012	Fernandes et al.	N/A	N/A
8596529	12/2012	Kolls	N/A	N/A
8600899	12/2012	Davis	N/A	N/A
8606702	12/2012	Ruckart	N/A	N/A
8615445	12/2012	Dorsey et al.	N/A	N/A
8645971	12/2013	Carlson et al.	N/A	N/A
8700530	12/2013	Smith	N/A	N/A
8707276	12/2013	Hill et al.	N/A	N/A
8712893	12/2013	Brandmaier	N/A	N/A

8761809	12/2013	Faith et al.	N/A	N/A
8769643	12/2013	Ben Ayed	N/A	N/A
8788341	12/2013	Patel	N/A	N/A
8794734	12/2013	Drummond	N/A	N/A
8810430	12/2013	Proud	N/A	N/A
8819659	12/2013	Ramer et al.	N/A	N/A
8831677	12/2013	Villa-Real	N/A	N/A
8838481	12/2013	Moshfeghi	N/A	N/A
8840016	12/2013	Schott et al.	N/A	N/A
8850421	12/2013	Proud	N/A	N/A
8856045	12/2013	Patel et al.	N/A	N/A
8881975	12/2013	Matthews	N/A	N/A
8898620	12/2013	Eizenman et al.	N/A	N/A
8903737	12/2013	Cameron et al.	N/A	N/A
8958846	12/2014	Freeny, Jr.	N/A	N/A
8976947	12/2014	Joyce et al.	N/A	N/A
9001047	12/2014	Forstall	N/A	N/A
9037492	12/2014	White	N/A	N/A
9047639	12/2014	Quintilian et al.	N/A	N/A
9092768	12/2014	Breitenbach et al.	N/A	N/A
9098961	12/2014	Block et al.	N/A	N/A
9210247	12/2014	Vance et al.	N/A	N/A
9262771	12/2015	Patel	N/A	N/A
9272713	12/2015	Dvoskin et al.	N/A	N/A
9395888	12/2015	Schiplacoff et al.	N/A	N/A
9424603	12/2015	Hammad	N/A	N/A
9483763	12/2015	Van Os et al.	N/A	N/A
9547859	12/2016	Patel	N/A	N/A
9659296	12/2016	Patel	N/A	N/A
9875473	12/2017	Patel	N/A	N/A
9898884	12/2017	Arora et al.	N/A	N/A
9911290	12/2017	Zalewski et al.	N/A	N/A
9922370	12/2017	Kobres et al.	N/A	N/A
10121318	12/2017	LeMay et al.	N/A	N/A
10163292	12/2017	Romero	N/A	N/A
10210501	12/2018	Low et al.	N/A	N/A
10217151	12/2018	Greiner et al.	N/A	N/A
10304057	12/2018	Powell	N/A	N/A
10380573	12/2018	Lin et al.	N/A	N/A
10410194	12/2018	Grassadonia	N/A	N/A
10423949	12/2018	Lyons et al.	N/A	N/A
10438208	12/2018	Patel et al.	N/A	N/A
10467604	12/2018	Dorsch et al.	N/A	N/A
10528944	12/2019	Khan	N/A	N/A
10538220	12/2019	Tyagi	N/A	B60R 25/245
10607465	12/2019	Subramanian et al.	N/A	N/A
10719833	12/2019	Patel et al.	N/A	N/A
10824828	12/2019	Ostri	N/A	N/A
10891608	12/2020	Patel	N/A	N/A

10891614	12/2020	Patel	N/A	N/A
10964185	12/2020	Subramanian et al.	N/A	N/A
10977642	12/2020	Khan	N/A	N/A
11010759	12/2020	Maeng	N/A	N/A
11042852	12/2020	Wadhwa	N/A	N/A
11074577	12/2020	Soccorsy et al.	N/A	N/A
11074580	12/2020	Patel	N/A	N/A
11182794	12/2020	Aument	N/A	N/A
11227275	12/2021	Van Heerden et al.	N/A	N/A
11308462	12/2021	Berman et al.	N/A	N/A
11373147	12/2021	Moore	N/A	N/A
11481772	12/2021	Patel	N/A	N/A
11488174	12/2021	Patel	N/A	N/A
11495104	12/2021	Subramanian et al.	N/A	N/A
11501296	12/2021	Patel	N/A	N/A
11564266	12/2022	Kahn	N/A	N/A
11935051	12/2023	Patel	N/A	N/A
11938692	12/2023	Hockings	N/A	N/A
11961107	12/2023	Patel	N/A	N/A
11966895	12/2023	Patel et al.	N/A	N/A
11966898	12/2023	Patel	N/A	N/A
11966920	12/2023	Patel	N/A	N/A
11966926	12/2023	Patel et al.	N/A	N/A
11972423	12/2023	Patel	N/A	N/A
12008875	12/2023	Subramanian et al.	N/A	N/A
12086811	12/2023	Patel	N/A	N/A
12093962	12/2023	Patel	N/A	N/A
12093963	12/2023	Patel	N/A	N/A
12124881	12/2023	Talavera	N/A	N/A
2002/0016740	12/2001	Ogasawara	N/A	N/A
2002/0152123	12/2001	Giordano et al.	N/A	N/A
2002/0164953	12/2001	Curtis	N/A	N/A
2003/0009385	12/2002	Tucciarone	N/A	N/A
2003/0089767	12/2002	Kiyomatsu	N/A	N/A
2003/0101096	12/2002	Suzuki et al.	N/A	N/A
2003/0110097	12/2002	Lei	N/A	N/A
2003/0130902	12/2002	Athwal	N/A	N/A
2003/0158891	12/2002	Lei et al.	N/A	N/A
2003/0172028	12/2002	Abell et al.	N/A	N/A
2003/0191811	12/2002	Hashem	N/A	N/A
2003/0206542	12/2002	Holder	N/A	N/A
2003/0236872	12/2002	Atkinson	N/A	N/A
2004/0029569	12/2003	Khan et al.	N/A	N/A
2004/0049454	12/2003	Kanno et al.	N/A	N/A
2004/0117262	12/2003	Berger et al.	N/A	N/A
2004/0122685	12/2003	Bunce et al.	N/A	N/A
2004/0133653	12/2003	Defosse	N/A	N/A
2005/0021459	12/2004	Bell	N/A	N/A
2005/0043011	12/2004	Murray	N/A	N/A
2005/0080510	12/2004	Bates	N/A	N/A

2005/0101295	12/2004	Rupp	N/A	N/A
2005/0177798	12/2004	Thomson et al.	N/A	N/A
2005/0181804	12/2004	Misikangas et al.	N/A	N/A
2005/0232421	12/2004	Simons et al.	N/A	N/A
2005/0234776	12/2004	Jacoves	N/A	N/A
2006/0043175	12/2005	Fu et al.	N/A	N/A
2006/0052157	12/2005	Walker et al.	N/A	N/A
2006/0123335	12/2005	Sanchez et al.	N/A	N/A
2007/0050083	12/2006	Signorelli et al.	N/A	N/A
2007/0083287	12/2006	Defosse et al.	N/A	N/A
2007/0095901	12/2006	Illingworth	N/A	N/A
2007/0119680	12/2006	Saltsov et al.	N/A	N/A
2007/0159994	12/2006	Brown et al.	N/A	N/A
2007/0186105	12/2006	Bailey et al.	N/A	N/A
2007/0187491	12/2006	Godwin et al.	N/A	N/A
2007/0227856	12/2006	Gopel	N/A	N/A
2007/0250920	12/2006	Lindsay	N/A	N/A
2007/0255653	12/2006	Tumminaro	N/A	N/A
2008/0010190	12/2007	Rackley, III et al.	N/A	N/A
2008/0010193	12/2007	Rackley, III et al.	N/A	N/A
2008/0033880	12/2007	Fiebiger et al.	N/A	N/A
2008/0040265	12/2007	Rackley, III et al.	N/A	N/A
2008/0126213	12/2007	Robertson et al.	N/A	N/A
2008/0141033	12/2007	Ginter et al.	N/A	N/A
2008/0154727	12/2007	Carlson	N/A	N/A
2008/0154735	12/2007	Carlson	N/A	N/A
2008/0163257	12/2007	Carlson et al.	N/A	N/A
2008/0167017	12/2007	Wentker et al.	N/A	N/A
2008/0167991	12/2007	Carlson et al.	N/A	N/A
2008/0183480	12/2007	Carlson et al.	N/A	N/A
2008/0201226	12/2007	Carlson et al.	N/A	N/A
2008/0208762	12/2007	Arthur et al.	N/A	N/A
2008/0249658	12/2007	Walker	N/A	N/A
2008/0254853	12/2007	Wright et al.	N/A	N/A
2008/0255947	12/2007	Friedman	N/A	N/A
2008/0319913	12/2007	Wiechers	N/A	N/A
2009/0037284	12/2008	Lewis et al.	N/A	N/A
2009/0076896	12/2008	Dewitt	N/A	N/A
2009/0099961	12/2008	Ogilvy	N/A	N/A
2009/0106160	12/2008	Skowronek	N/A	N/A
2009/0119190	12/2008	Realini	N/A	N/A
2009/0157515	12/2008	LaFauci et al.	N/A	N/A
2009/0171682	12/2008	Dixon et al.	N/A	N/A
2009/0287349	12/2008	Mardiks	N/A	N/A
2009/0288173	12/2008	Mardiks	N/A	N/A
2009/0303982	12/2008	Blachman et al.	N/A	N/A
2009/0306818	12/2008	Slagley et al.	N/A	N/A
2009/0306819	12/2008	Insolia	N/A	N/A
2009/0313125	12/2008	Roh et al.	N/A	N/A
2009/0313132	12/2008	Kenna et al.	N/A	N/A

2009/0327089	12/2008	Kanno et al.	N/A	N/A
2010/0061294	12/2009	Proctor, Jr.	N/A	N/A
2010/0082485	12/2009	Lin et al.	N/A	N/A
2010/0094456	12/2009	Simpkins et al.	N/A	N/A
2010/0105454	12/2009	Weber et al.	N/A	N/A
2010/0198400	12/2009	Pascal	N/A	N/A
2010/0227671	12/2009	Laaroussi et al.	N/A	N/A
2010/0276484	12/2009	Banerjee	N/A	N/A
2010/0280956	12/2009	Chutorash	N/A	N/A
2010/0312692	12/2009	Teicher	N/A	N/A
2010/0320266	12/2009	White	N/A	N/A
2010/0329285	12/2009	Stanton	N/A	N/A
2011/0029405	12/2010	Cronin	N/A	N/A
2011/0040686	12/2010	Carlson	N/A	N/A
2011/0125561	12/2010	Marcus	N/A	N/A
2011/0153436	12/2010	Krampe	N/A	N/A
2011/0153442	12/2010	Krampe	N/A	N/A
2011/0153495	12/2010	Dixon et al.	N/A	N/A
2011/0172848	12/2010	Breitenbach et al.	N/A	N/A
2011/0178883	12/2010	Granbery	N/A	N/A
2011/0225067	12/2010	Dunwoody	N/A	N/A
2011/0238476	12/2010	Carr	N/A	N/A
2011/0244799	12/2010	Roberts et al.	N/A	N/A
2011/0251892	12/2010	Laracey	N/A	N/A
2011/0251910	12/2010	Dimmick	N/A	N/A
2011/0276636	12/2010	Cheng et al.	N/A	N/A
2011/0289023	12/2010	Forster et al.	N/A	N/A
2012/0011024	12/2011	Dorsey et al.	N/A	N/A
2012/0016731	12/2011	Smith et al.	N/A	N/A
2012/0029691	12/2011	Mockus et al.	N/A	N/A
2012/0030047	12/2011	Fuentes	N/A	N/A
2012/0036045	12/2011	Lowe et al.	N/A	N/A
2012/0066096	12/2011	Penide	N/A	N/A
2012/0078735	12/2011	Bauer et al.	N/A	N/A
2012/0108173	12/2011	Hahm et al.	N/A	N/A
2012/0122585	12/2011	Nguyen	N/A	N/A
2012/0136478	12/2011	Anand	N/A	N/A
2012/0150742	12/2011	Poon et al.	N/A	N/A
2012/0158172	12/2011	Wencslao	N/A	N/A
2012/0158528	12/2011	Hsu et al.	N/A	N/A
2012/0160912	12/2011	Laracey	N/A	N/A
2012/0173325	12/2011	Johri	N/A	N/A
2012/0197740	12/2011	Grigg et al.	N/A	N/A
2012/0203666	12/2011	Torossian et al.	N/A	N/A
2012/0231844	12/2011	Coppinger	N/A	N/A
2012/0246074	12/2011	Annamalai et al.	N/A	N/A
2012/0253852	12/2011	Pourfallah	N/A	N/A
2012/0254631	12/2011	Skillman et al.	N/A	N/A
2012/0255653	12/2011	Chin	N/A	N/A
2012/0258773	12/2011	Alvarez Rivera	N/A	N/A

2012/0276845	12/2011	Wikander	N/A	N/A
2012/0290472	12/2011	Mullen et al.	N/A	N/A
2012/0296826	12/2011	Bergdale et al.	N/A	N/A
2012/0303528	12/2011	Weiner et al.	N/A	N/A
2012/0316963	12/2011	Moshfeghi	N/A	N/A
2012/0330764	12/2011	Nahidipour	N/A	N/A
2012/0330769	12/2011	Arceo	N/A	N/A
2012/0330844	12/2011	Kaufman	N/A	N/A
2013/0030931	12/2012	Moshfeghi	N/A	N/A
2013/0054016	12/2012	Canter et al.	N/A	N/A
2013/0054336	12/2012	Graylin	N/A	N/A
2013/0054395	12/2012	Cyr et al.	N/A	N/A
2013/0067365	12/2012	Shrufi et al.	N/A	N/A
2013/0085835	12/2012	Horowitz	N/A	N/A
2013/0087050	12/2012	Studor et al.	N/A	N/A
2013/0100886	12/2012	Cherian	N/A	N/A
2013/0102246	12/2012	Gagne	455/41.1	G06Q 20/36
2013/0110296	12/2012	Khoo	N/A	N/A
2013/0117490	12/2012	Harriman	N/A	N/A
2013/0117738	12/2012	Livingston et al.	N/A	N/A
2013/0124289	12/2012	Fisher	N/A	N/A
2013/0126607	12/2012	Behjat	N/A	N/A
2013/0143498	12/2012	Niemi	N/A	N/A
2013/0166448	12/2012	Narayanan	N/A	N/A
2013/0185150	12/2012	Crum	N/A	N/A
2013/0191213	12/2012	Beck et al.	N/A	N/A
2013/0191789	12/2012	Calman	N/A	N/A
2013/0217333	12/2012	Sprigg et al.	N/A	N/A
2013/0246171	12/2012	Carapelli	N/A	N/A
2013/0246364	12/2012	Bhavith	N/A	N/A
2013/0267121	12/2012	Hsu	N/A	N/A
2013/0267176	12/2012	Hertel et al.	N/A	N/A
2013/0275303	12/2012	Fiore	N/A	N/A
2013/0275305	12/2012	Duplan	N/A	N/A
2013/0278622	12/2012	Sun et al.	N/A	N/A
2013/0282590	12/2012	Rajarethnam et al.	N/A	N/A
2013/0284806	12/2012	Margalit	N/A	N/A
2013/0297422	12/2012	Hunter et al.	N/A	N/A
2013/0311379	12/2012	Smith	N/A	N/A
2013/0311382	12/2012	Fosmark et al.	N/A	N/A
2013/0331985	12/2012	Felique	N/A	N/A
2013/0332293	12/2012	Ran	N/A	N/A
2013/0346302	12/2012	Purves et al.	N/A	N/A
2013/0346305	12/2012	Mendes	N/A	N/A
2014/0006451	12/2013	Mullis et al.	N/A	N/A
2014/0012414	12/2013	Pérez et al.	N/A	N/A
2014/0019367	12/2013	Khan et al.	N/A	N/A
2014/0025958	12/2013	Calman	N/A	N/A
2014/0032410	12/2013	Georgiev et al.	N/A	N/A

2014/0032413	12/2013	Low	N/A	N/A
2014/0040028	12/2013	King et al.	N/A	N/A
2014/0040117	12/2013	Jain	N/A	N/A
2014/0046842	12/2013	Irudayam et al.	N/A	N/A
2014/0052463	12/2013	Cashman et al.	N/A	N/A
2014/0052524	12/2013	Andersen	N/A	N/A
2014/0052607	12/2013	Park	N/A	N/A
2014/0064116	12/2013	Linde et al.	N/A	N/A
2014/0067542	12/2013	Everingham	N/A	N/A
2014/0074714	12/2013	Melone et al.	N/A	N/A
2014/0074723	12/2013	Kamat	N/A	N/A
2014/0085046	12/2013	Shin et al.	N/A	N/A
2014/0085109	12/2013	Stefik	N/A	N/A
2014/0089016	12/2013	Smullin	N/A	N/A
2014/0089185	12/2013	Desai	705/41	G06Q 10/067
2014/0100977	12/2013	Davis	N/A	N/A
2014/0108108	12/2013	Artman	N/A	N/A
2014/0122298	12/2013	Oyer	N/A	N/A
2014/0136301	12/2013	Valdes	N/A	N/A
2014/0136411	12/2013	Cho	N/A	N/A
2014/0143055	12/2013	Johnson	N/A	N/A
2014/0143074	12/2013	Kolls	N/A	N/A
2014/0143137	12/2013	Carlson	N/A	N/A
2014/0172179	12/2013	Baudin	N/A	N/A
2014/0180852	12/2013	Kamat	N/A	N/A
2014/0188708	12/2013	Govindarajan et al.	N/A	N/A
2014/0201066	12/2013	Roux et al.	N/A	N/A
2014/0249995	12/2013	Ogilvy	N/A	N/A
2014/0278989	12/2013	Calman	N/A	N/A
2014/0279008	12/2013	Calman	N/A	N/A
2014/0279101	12/2013	Duplan et al.	N/A	N/A
2014/0279426	12/2013	Holman et al.	N/A	N/A
2014/0279537	12/2013	Cicoretti	N/A	N/A
2014/0279556	12/2013	Priebatsch	N/A	N/A
2014/0289047	12/2013	Yee	N/A	N/A
2014/0317611	12/2013	Wojcik et al.	N/A	N/A
2014/0324627	12/2013	Haver	N/A	N/A
2014/0337235	12/2013	Van Heerden et al.	N/A	N/A
2014/0351099	12/2013	Zhu	N/A	N/A
2014/0361872	12/2013	Garcia et al.	N/A	N/A
2014/0378057	12/2013	Ramon et al.	N/A	N/A
2015/0006421	12/2014	Pearson	N/A	N/A
2015/0019433	12/2014	Leger	N/A	N/A
2015/0051977	12/2014	Lyman	N/A	N/A
2015/0073980	12/2014	Griffin et al.	N/A	N/A
2015/0081462	12/2014	Ozvat et al.	N/A	N/A
2015/0088698	12/2014	Ackerman	N/A	N/A
2015/0100152	12/2014	Trevino et al.	N/A	N/A
2015/0105901	12/2014	Joshi et al.	N/A	N/A

2015/0120546	12/2014	Fernandes	N/A	N/A
2015/0120555	12/2014	Jung	N/A	N/A
2015/0149992	12/2014	Wade et al.	N/A	N/A
2015/0154579	12/2014	Teicher	N/A	N/A
2015/0169312	12/2014	Patel	N/A	N/A
2015/0170131	12/2014	Patel	N/A	N/A
2015/0170132	12/2014	Patel	N/A	N/A
2015/0170136	12/2014	Patel	N/A	N/A
2015/0178702	12/2014	Patel	N/A	N/A
2015/0206128	12/2014	Torossian et al.	N/A	N/A
2015/0220381	12/2014	Horagan et al.	N/A	N/A
2015/0235202	12/2014	Zabala	N/A	N/A
2015/0278811	12/2014	Lalchandani	N/A	N/A
2015/0287085	12/2014	Windmueller	N/A	N/A
2015/0302377	12/2014	Sweitzer	N/A	N/A
2015/0302411	12/2014	Bondesen et al.	N/A	N/A
2015/0317720	12/2014	Ramaratnam	N/A	N/A
2015/0332029	12/2014	Coxe	N/A	N/A
2015/0346994	12/2014	Chanyontpatanakul	N/A	N/A
2015/0373537	12/2014	Toksvig	N/A	N/A
2015/0379491	12/2014	Ma et al.	N/A	N/A
2016/0012465	12/2015	Sharp	N/A	N/A
2016/0019604	12/2015	Kobayashi	N/A	N/A
2016/0063476	12/2015	Baldie	N/A	N/A
2016/0086145	12/2015	Tsutsui	N/A	N/A
2016/0092859	12/2015	Klingen	N/A	N/A
2016/0098690	12/2015	Silvia et al.	N/A	N/A
2016/0132870	12/2015	Xu et al.	N/A	N/A
2016/0196220	12/2015	Perez et al.	N/A	N/A
2016/0232515	12/2015	Jhas	N/A	N/A
2016/0260081	12/2015	Zermeño	N/A	N/A
2016/0292469	12/2015	Ianni	N/A	N/A
2016/0335620	12/2015	Lyons et al.	N/A	N/A
2016/0350744	12/2015	Tang et al.	N/A	N/A
2017/0006656	12/2016	Nacer et al.	N/A	N/A
2017/0017958	12/2016	Scott et al.	N/A	N/A
2017/0083882	12/2016	Kim	N/A	G06Q 20/102
2017/0092062	12/2016	Tsutsui	N/A	N/A
2017/0148009	12/2016	Lafuente	N/A	N/A
2017/0161728	12/2016	Satyanarayan et al.	N/A	N/A
2017/0178116	12/2016	Savolainen et al.	N/A	N/A
2017/0193478	12/2016	Dhurka	N/A	N/A
2017/0193479	12/2016	Kamat	N/A	N/A
2017/0193508	12/2016	Patel et al.	N/A	N/A
2017/0330164	12/2016	Suelberg et al.	N/A	N/A
2018/0005220	12/2017	Laracey	N/A	N/A
2018/0096323	12/2017	Barber et al.	N/A	N/A
2018/0096554	12/2017	Hough et al.	N/A	N/A
2018/0165908	12/2017	Patel et al.	N/A	N/A

2018/0197167	12/2017	Ganesan et al.	N/A	N/A
2018/0240096	12/2017	Patel	N/A	N/A
2018/0276674	12/2017	Ramatchandirane et al.	N/A	N/A
2018/0293573	12/2017	Ortiz et al.	N/A	N/A
2018/0315271	12/2017	Gharabegian et al.	N/A	N/A
2018/0374076	12/2017	Wheeler et al.	N/A	N/A
2019/0213832	12/2018	Tsutsui	N/A	N/A
2019/0236586	12/2018	Mei et al.	N/A	N/A
2019/0244205	12/2018	Fieglein	N/A	N/A
2019/0244465	12/2018	Saunders et al.	N/A	N/A
2019/0272539	12/2018	Bogaard	N/A	N/A
2019/0340604	12/2018	Nadella et al.	N/A	N/A
2020/0126059	12/2019	Kudssi et al.	N/A	N/A
2020/0250948	12/2019	Ball	N/A	N/A
2020/0342507	12/2019	Hubbard et al.	N/A	N/A
2020/0387881	12/2019	Smith et al.	N/A	N/A
2021/0012318	12/2020	Ducoulombier	N/A	N/A
2021/0056552	12/2020	Murray	N/A	N/A
2021/0158309	12/2020	Mcginlay et al.	N/A	N/A
2021/0158322	12/2020	Cohn et al.	N/A	N/A
2021/0192479	12/2020	Swaminathan	N/A	N/A
2021/0209671	12/2020	Barela et al.	N/A	N/A
2021/0312440	12/2020	Badalian et al.	N/A	N/A
2021/0334925	12/2020	Peacock et al.	N/A	N/A
2021/0357932	12/2020	Patel	N/A	N/A
2021/0375094	12/2020	Thomas et al.	N/A	N/A
2021/0406970	12/2020	Patel et al.	N/A	N/A
2022/0172549	12/2021	Chung et al.	N/A	N/A
2022/0263886	12/2021	Rose	N/A	N/A
2022/0292543	12/2021	Henderson	N/A	N/A
2022/0405733	12/2021	Yao et al.	N/A	N/A
2023/0042066	12/2022	Liff et al.	N/A	N/A
2023/0222506	12/2022	Patel et al.	N/A	N/A
2023/0222507	12/2022	Patel et al.	N/A	N/A
2023/0274274	12/2022	Patel	N/A	N/A
2023/0281621	12/2022	Patel	N/A	N/A
2023/0289811	12/2022	Patel et al.	N/A	N/A
2023/0297987	12/2022	Patel	N/A	N/A
2024/0257093	12/2023	Wulff et al.	N/A	N/A
2024/0273497	12/2023	Patel et al.	N/A	N/A
2024/0273498	12/2023	Patel	N/A	N/A
2024/0273526	12/2023	Patel	N/A	N/A
2024/0273527	12/2023	Patel	N/A	N/A
2024/0273528	12/2023	Patel	N/A	N/A
2024/0273541	12/2023	Patel	N/A	N/A
2024/0330974	12/2023	Patel	N/A	N/A
2024/0338693	12/2023	Patel	N/A	N/A
2024/0428254	12/2023	Patel et al.	N/A	N/A
2025/0014044	12/2024	Patel et al.	N/A	N/A

2025/0029108	12/2024	Patel	N/A	N/A
2025/0094977	12/2024	Patel et al.	N/A	N/A
2025/0148891	12/2024	Patel	N/A	N/A
2025/0209467	12/2024	Patel et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2015327722	12/2016	AU	N/A
2021107597	12/2021	AU	N/A
3028753	12/2018	CA	N/A
2844231	12/2019	CA	N/A
1561508	12/2004	CN	N/A
103688252	12/2013	CN	N/A
104050747	12/2013	CN	N/A
106104600	12/2014	CN	N/A
104603809	12/2014	CN	N/A
204375056	12/2014	CN	N/A
105139196	12/2014	CN	N/A
105989466	12/2015	CN	N/A
106803175	12/2016	CN	N/A
107480975	12/2016	CN	N/A
104680361	12/2017	CN	N/A
207663510	12/2017	CN	N/A
108367497	12/2017	CN	N/A
109118199	12/2018	CN	N/A
109389755	12/2018	CN	N/A
111316301	12/2019	CN	N/A
107408253	12/2020	CN	N/A
108352094	12/2020	CN	N/A
115485782	12/2021	CN	N/A
113379401	12/2023	CN	N/A
1571607	12/2004	EP	N/A
2061001	12/2008	EP	N/A
3901880	12/2020	EP	N/A
3289548	12/2020	EP	N/A
2504922	12/2013	GB	N/A
H1125320	12/1998	JP	N/A
2002-183812	12/2001	JP	N/A
2003-242401	12/2002	JP	N/A
2003-323662	12/2002	JP	N/A
2004-252640	12/2003	JP	N/A
2004310740	12/2003	JP	N/A
WO2003017157	12/2003	JP	N/A
2005-526325	12/2004	JP	N/A
2009-259226	12/2008	JP	N/A
2010528716	12/2009	JP	N/A
4586607	12/2009	JP	N/A
2012-504273	12/2011	JP	N/A
7289565	12/2022	JP	N/A

20130138637	12/2012	KR	N/A
20160076081	12/2015	KR	N/A
20170140189	12/2016	KR	N/A
WO2003/098561	12/2002	WO	N/A
WO2006/020692	12/2005	WO	N/A
WO2007/015610	12/2006	WO	N/A
WO2008/083022	12/2007	WO	N/A
WO2008/083025	12/2007	WO	N/A
WO2008/083078	12/2007	WO	N/A
WO2008/083089	12/2007	WO	N/A
WO2008/083105	12/2007	WO	N/A
WO2008/083115	12/2007	WO	N/A
WO2008/083119	12/2007	WO	N/A
WO2009/070430	12/2008	WO	N/A
WO2009089163	12/2008	WO	N/A
WO2013/132995	12/2012	WO	N/A
WO2013/177416	12/2012	WO	N/A
WO2014/093857	12/2013	WO	N/A
WO2015123378	12/2014	WO	N/A
WO2015186141	12/2014	WO	N/A
WO2016/123545	12/2015	WO	N/A
WO-2016137271	12/2015	WO	G06F 3/041
WO2016158748	12/2015	WO	N/A
WO2017/010936	12/2016	WO	N/A
WO2017/143079	12/2016	WO	N/A
WO2023037112	12/2022	WO	N/A

OTHER PUBLICATIONS

US 12,125,030 B2, 10/2024, Patel (withdrawn) cited by applicant

Goetting "Device Description HG G-76343/4/5", Jan. 24, 2024, 130 pages (Year: 2024). cited by examiner

@RobocopyEs, posted Oct. 11, 2014, retrieved Feb. 13, 2018,

<URL:https://twitter.com/robocopyes> 2 pgs. cited by applicant

Adams, How can stationary kiosks thrive in a mobile world?, American Banker, 2012. cited by applicant

Balan et al., mFerio: the design and evaluation of a peer-to-peer mobile payment system, Jun. 2009, 14 pgs. cited by applicant

Balfe et al., e-EMV: emulating EMV for internet payments with trusted computing Technologies, Oct. 2008, 12 pgs. cited by applicant

Bing, Bing Images Search: "dongle", <http://www.bing.com/images/search?q=dongleFORM+HDRSC2>, Dec. 5, 2013, 8 pgs. cited by applicant

Carlson, Specification, U.S. Appl. No. 60/871,898, filed Dec. 26, 2006, 169 pgs. cited by applicant

Frolick, Assessing M-Commerce Opportunities, Auerbach Publications Inc., Information Systems Management, Spring 2004. cited by applicant

Google, Chromecast, <http://www.google.com/intl/devices/chromecast/>, Dec. 12, 2013, 4 pgs. cited by applicant

How to Pay the New Way, YouTube, Apr. 5, 2018, 4 pgs. cited by applicant

How will Apple's new mobile wallet Passbook impact other mobile wallets?, posted Jun. 13, 2012, retrieved Feb. 13, 2018 from <URL:https://www.quora.com/How-will-Apples-new-mobile-wallet-Passbook-impact-other-mobile-wallets>, 5 pgs. cited by applicant

Kadambi et al., Near-Field Communication-based Secure Mobile Payment Service, Aug. 2009, 10 pgs. cited by applicant

When the Future Feels Worse Than the Past: A Temporal Inconsistency in Moral Judgment, 15 pgs. (Year: 2010) <https://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.675.3584&rep=rep1&type=pdf>. cited by applicant

Novotny, Applying RFID technology in the retail industry-benefits and concerns from the consumer's perspective, Institute of Economic Science, Eszterhazy Karoly College, Eger, Hungary, Retail Technologies for the 21 Century, innovation and competitiveness in the retail industry, 2015. cited by applicant

Nurel, "Recent Developments in Wireless Network Systems", Izmir Institute of Technology, Sep. 2001, 280 pages (Year: 2001). cited by applicant

Patel, Office Action, U.S. Appl. No. 14/320,534, Mar. 2, 2018, 26 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/320,534, Apr. 16, 2015, 21 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/320,534, Nov. 30, 2016, 24 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/321,717, Jun. 18, 2015, 22 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/321,724, Oct. 8, 2015, 19 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/321,724, Dec. 13, 2017, 22 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/321,733, Nov. 14, 2014, 11 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/335,762, Jun. 9, 2016, 15 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/456,683, Jun. 8, 2015, 14 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/458,192, Sep. 16, 2015, 26 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/458,199, Jun. 24, 2015, 8 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/641,236, Mar. 11, 2016, 16 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 14/968,703, Feb. 12, 2019, 22 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 15/435,228, Oct. 2, 2020, 24 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 15/893,514, Jul. 22, 2021, 12 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 15/956,741, Oct. 2, 2020, 12 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/214,644, Jun. 10, 2014, 9 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/321,733, Jun. 22, 2015, 8 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/321,733, Feb. 27, 2015, 9 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/335,762, Oct. 3, 2016, 7 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/335,762, Mar. 30, 2015, 9 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/456,683, Oct. 8, 2015, 15 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/458,192, Oct. 12, 2017, 8 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/458,199, Jan. 20, 2017, 9 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/611,065, Mar. 26, 2018, 18 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/614,336, Dec. 11, 2015, 8 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/614,336, Nov. 25, 2015, 13 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 14/968,703, Jun. 27, 2019, 10 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 15/406,492, Mar. 11, 2020, 10 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 15/435,228, Aug. 12, 2021, 9 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 15/603,400, Dec. 18, 2019, 9 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 15/603,400, Jun. 18, 2020, 5 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 15/878,352, Oct. 23, 2020, 9 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 16/029,483, Dec. 23, 2020, 23 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 16/748,727, May 9, 2022, 18 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 16/748,727, Jan. 20, 2022, 17 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 16/750,477, Jan. 26, 2022, 17 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 16/934,933, Mar. 31, 2021, 9 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 16/681,673, Aug. 17, 2022, 22 pgs. cited by applicant

Patel, Notice of Allowability, U.S. Appl. No. 16/934,392, Sep. 28, 2022, 2 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/529,111, Sep. 22, 2022, 10 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/654,732, Sep. 16, 2022, 9 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/320,534, Apr. 8, 2016, 21 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/320,534, Oct. 29, 2014, 18 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/321,717, Dec. 19, 2014, 16 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/321,724, Mar. 13, 2017, 21 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/321,724, May 15, 2015, 19 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/321,733, Aug. 21, 2014, 9 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/335,762, Dec. 10, 2014, 7 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/335,762, Sep. 18, 2015, 13 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/456,683, Jan. 2, 2015, 10 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/458,192, Mar. 23, 2017, 26 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/458,192, Jan. 30, 2015, 24 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/458,199, Jan. 5, 2015, 7 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/458,199, Mar. 28, 2016, 8 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/611,065, Oct. 3, 2016, 19 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/611,065, Jun. 13, 2017, 17 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/614,336, May 27, 2015, 17 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/641,236, Feb. 7, 2018, 19 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/641,236, May 29, 2015, 10 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 14/968,703, Aug. 7, 2018, 31 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 15/406,492, Ju. 25, 2019, 17 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 15/435,228, Mar. 26, 2020, 21 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 15/603,400, Jun. 12, 2019, 11 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 15/878,352, Jan. 24, 2020, 13 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 15/893,514, Oct. 29, 2020, 17 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 15/956,741, Apr. 22, 2020, 10 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 15/956,741, Dec. 27, 2021, 10 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 16/029,483, Apr. 27, 2020, 28 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 16/681,673, Dec. 24, 2021, 21 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 16/934,933, Oct. 28, 2020, 10 pgs. cited by applicant

applicant
Patel, Non-Final Office Action, U.S. Appl. No. 17/216,399, Apr. 8, 2022, 15 pgs. cited by applicant
Patel, Non-Final Office Action, U.S. Appl. No. 15/893,514, Sep. 30, 2022, 8 pgs. cited by applicant
PayRange Inc., Communication Pursuant to Article 94(3), EP14828617.2, Dec. 19, 2017, 6 pgs.
cited by applicant
PayRange Inc., Communication Pursuant to Article 94(3), EP16706931.9, Jun. 29, 2018, 8 pgs.
cited by applicant
PayRange Inc., Communication Pursuant to Rules 161(1) and 162, EP148286172, Sep. 21, 2016, 2
pgs. cited by applicant
PayRange Inc., Communication Pursuant to Rules 161(1) and 162, EP16706931.9, Sep. 21, 2017, 2
pgs. cited by applicant
PayRange Inc., Communication under Rule 71(3) EPC, EP14828617.2, Nov. 19, 2020, 7 pgs. cited
by applicant
PayRange Inc., Communication under Rule 71(3) EPC, EP17708929.9, Jun. 12, 2020, 7 pgs. cited
by applicant
PayRange Inc., European Search Report, EP20203134.0, Mar. 1, 2021, 7 pgs. cited by applicant
PayRange Inc., European Search Report, EP21165692.1, Sep. 14, 2021, 10 pgs. cited by applicant
PayRange Inc., International Preliminary Report on Patentability, PCT/US2014/071284, Jun. 21,
2016, 6 pgs. cited by applicant
PayRange Inc., International Preliminary Report on Patentability, PCT/US2016/015763, Aug. 1,
2017, 7 pgs. cited by applicant
PayRange Inc., International Preliminary Report on Patentability, PCT/US2017/015676, Jul. 31,
2018, 9 pgs. cited by applicant
PayRange Inc., International Preliminary Report on Patentability, PCT/US2017/018194, Aug. 21,
2018, 17 pgs. cited by applicant
PayRange Inc., International Preliminary Report on Patentability, PCT/US2019/060777, May 11,
2021, 7 pgs. cited by applicant
PayRange Inc., International Search Report and Written Opinion, PCT/US2014/071284, Mar. 25,
2015, 9 pgs. cited by applicant
PayRange Inc., International Search Report and Written Opinion, PCT/US2016/015763, Apr. 8,
2016, 9 pgs. cited by applicant
PayRange Inc., International Search Report and Written Opinion, PCT/US2017/015676, Apr. 18,
2017, 11 pgs. cited by applicant
PayRange Inc., International Search Report and Written Opinion, PCT/US2017/018194, Apr. 12,
2017, 10 pgs. cited by applicant
PayRange Inc., International Search Report and Written Opinion, PCT/US2019/060777, Feb. 6,
2020, 11 pgs. cited by applicant
PayRange Inc., International Search Report and Written Opinion, PCT/US2021/042632, Nov. 17,
2021, 11 pgs. cited by applicant
PayRange Inc., Notice of Reasons for Rejection, JP2017527886, Aug. 29, 2019, 10 pgs. cited by
applicant
PayRange Inc., Notice of Reasons for Rejection, JP2018-543707, Sep. 4, 2020, 4 pgs. cited by
applicant
PayRange Inc., Notice of Reasons for Rejection, JP2020-101558, Oct. 7, 2021, 4 pgs. cited by
applicant
PayRange Inc., Summons to Attend Oral Proceedings, EP14828617.2, Apr. 2, 2020, 12 pgs. cited
by applicant
PayRange New Product Launch, posted at youtube.com Jun. 27, 2015, © 2016 YouTube, LLC,
[online], [site visited Mar. 2, 2016]. Available from Internet, <URL:
<https://www.youtube.com/watch?v=NTvvV03XFeg>, 1 pg. cited by applicant

Smart Vending Machine Demo at TechCrunch Disrupt 2013, posted at youtube.com Dec. 3, 2013, © 2016 YouTube, LLC, [online], [site visited Mar. 2, 2016]. Available from internet, URL: <https://www.youtube.com/watch?v=XEz1H-gxLj8>>. cited by applicant

Square Mobile Credit Card Processing for iPhone, iPod, iPad, posted at youtube.com, posting date Apr. 30, 2011, © 2016 YouTube, LLC, [online], [site visited Mar. 2, 2016]. Available from internet, <URL: <https://www.youtube.com/watch?v=v6sKb3CFSKw>>. cited by applicant

Kanapaka et al., A Stochastic Game Theoretic Model for Expanding ATM Services. <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=7395687>, 2015, 8 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/147,305, Oct. 6, 2022, 9 pgs. cited by applicant

Hoffman et al., “New options in Wireless payments”, Internet World 7.7:37 Penton Media Inc., Penton Business Media, Inc. and their subsidiaries. (Year: 2001) 5 pgs. cited by applicant

Carton et al., Framework for Mobile Payments Integration, Electronic Journal of Information Systems Evaluation, 15.1: 14-24, Academic Conferences International Limited, January. (Year: 2012), 14 pgs. cited by applicant

Apriva LLC Awarded Patent for System and Method for Facilitating a Purchase Transaction using a Customer Device Beacon, Jun. 7, 2017, Global IP News (Year: 2017), 5 pgs. cited by applicant

Kumar, “Amazon gets Indian patent for auto authentication of mobile transactions”, ProQuest document Id:2433007646, Financial Express, (Aug. 13, 2020), 2 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 17/443,802, Dec. 23, 2022, 14 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 15/956,741, Feb. 27, 2023, 11 pgs. cited by applicant

Patel et al., Notice of Allowance, U.S. Appl. No. 15/893,514, Apr. 10, 2023, 13 pgs. cited by applicant

Heimerl et al., “Community sourcing: Engaging Local Crowds to Perform Expert Work Via Physical Kiosks”, CHI '12: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, May 2012, pp. 1539-1548, 10 pgs. <https://doi.org/10.1145/2207676.2208619>. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/443,802, Jun. 28, 2023, 8 pgs. cited by applicant

Patel, Corrected Notice of Allowability, U.S. Appl. No. 17/443,802, Jul. 10, 2023, 5 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/983,311, Jun. 28, 2023, 10 pgs. cited by applicant

EIC 3600 Search Report, STIC, Scientific & Technical Information Center, Date Completed Jun. 12, 2023, 5 pgs. cited by applicant

Patel et al., Notice of Allowance, U.S. Appl. No. 15/893,514, Jul. 12, 2023, 13 pgs. cited by applicant

Patel et al., Notice of Allowance, U.S. Appl. No. 17/973,506, Jul. 26, 2023, 13 pgs. cited by applicant

Katy Jacob, “Are mobile payments the smart cards of the aughts?”, Scientific and Technical Information Center, Report Information from Dialog, Jul. 14, 2023—11:33, ProQuest, Publication Info: Chicago Fed Letter 240: 1-4. Federal Reserve Bank of Chicago. (Jul. 2007), 9 pgs. cited by applicant

Patel et al., Notice of Allowance, U.S. Appl. No. 17/963,170, Aug. 4, 2023, 16 pgs. cited by applicant

USA Technologies Announces Cashless Solution to Be Offered by Blackboard Inc., Scientific and Technical Information Center, Report Information from Dialog, Jul. 25, 2023, ProQuest, Publication Info: Business Wire Jul. 18, 2007: NA, 6 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 18/197,071, Aug. 16, 2023, 9 pgs. cited by applicant

Hossain et al., “Comprehensive Study of Bluetooth Signal Parameters for Localization”,

Department of Electrical & Computer Engineering National University of Singapore, 5 pgs. Email: {g0500774, weeseng}@nus.edu.sg. cited by applicant

Bluetooth Specification 1.2, Covered Core Package version: 1.2, Current Master TOC issued: Nov. 5, 2003, 1200 pgs. cited by applicant

Hands-Free Profile 1.5, Doc. No. HFP1.5_SPEC, Nov. 25, 2005, 93 pgs. cited by applicant

DEX and MDB: A Primer For Vendors | Vending Market Watch, Feb. 7, 2008, 5 pgs.
<https://www.vendingmarketwatch.com/print/content/10272928>. cited by applicant

MDB Protocol V4.2—Multi-Drop Bus—Internal Communication Protocol, MDB / ICP, Version 4.2, Feb. 2011, 313 pgs. cited by applicant

Gruber et al., “The Commodity Vending Machine”, Forum Ware International Feb. 2005, 11 pgs. cited by applicant

Michael L. Kasavana, Innovative VDI Standards: Moving an Industry Forward, The Journal of International Management, vol. 4, No. 3, Dec. 2009, 10 pgs. cited by applicant

SDFL Administrative Order 2021-33, Apr. 6, 2021, 5 pgs. cited by applicant

The New York Times by David Pogue, In Arrived of 2 iPhones, 3 Lessons, Sep. 17, 2013, 4 pgs.
<https://www.nytime.com/2013/09/18/technology/personaltech/In-Arrived-of-2-iPhones-3-Lessons.html>. cited by applicant

Cnet, John Thompson, How to use S Beam on your Samsung Galaxy S3, Jun. 21, 2012, 5 pgs.
<https://www.cnet.com/how-to/how-to-use-s-beam-on-your-samsung-galaxy-s3/>. cited by applicant

Iphone, User Guide For iOS 6.1 Software, 156 pgs. cited by applicant

Apple Reports Fourth Quarter Results, Oct. 28, 2013, 4 pgs. cited by applicant

Apple Announces iPhone 5s—The Most Forward—Thinking Smartphone in the World, Sep. 10, 2013, 5 pgs. cited by applicant

cNet, by Marguerite Reardon, Motion sensing comes to mobile phones, Jun. 11, 2007, 4 pgs. cited by applicant

Multi-Drop Bus—Internal Communication Protocol, MDB / ICP, Version 3, Mar. 26, 2003, 270 pgs. cited by applicant

Weidong Kou, Payment Technologies for E-Commerce, University of Hong Kong Pokfulam Road, Hong Kong, ACM Subject Classification (1998): H.4, K.4.4, J.1, 339 pgs. cited by applicant

Specification for RFID Air Interface, EPCTM Radio-Frequency Identity Protocols, Class-1 Generation-2 UHF RFID, Protocol for Communications at 860 MHz-960 MHz, Version 1.2.0, EPCglobal Inc., Oct. 23, 2008, 108 pgs. cited by applicant

Baier et al., “Principles of Model Checking”, The MIT Press Cambridge, Massachusetts, London, England, 2008, 994 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/983,311, Oct. 4, 2023, 11 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 18/197,070, Sep. 27, 2023, 8 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/443,802, Nov. 1, 2023, 8 pgs. cited by applicant

Patel et al., Notice of Allowance, U.S. Appl. No. 15/893,514, Nov. 8, 2023, 13 pgs. cited by applicant

Kevin Werbach et al., “Contracts Ex Machina”, Articles, Faculty Scholarship, University of Michigan Law School, The University of Michigan Law School Scholarship Repository, (Year: 2017), 71 pgs. <https://repository.law.umich.edu/cgi/viewcontent.cgi?article=2936&context=articles>. cited by applicant

Patel et al., Notice of Allowance, U.S. Appl. No. 17/973,506, Dec. 7, 2023, 11 pgs. cited by applicant

Bailey, Mobile-phone payment option gains ground, Dialog, STIC, Report Information from Dialog, Nov. 29, 2023, 6 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 17/968,672, Dec. 15, 2023, 14 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 17/973,505, Dec. 15, 2023, 14 pgs. cited by

applicant

Patel, Notice of Allowance, U.S. Appl. No. 18/197,070, Dec. 20, 2023, 10 pgs. cited by applicant
“Mobile payment heads to the U.S.”, Mobile Internet 6.10:1. Information Gatekeepers, Inc. (Oct. 2004), 2 pgs. cited by applicant

“During the period 2005 to 2009, the total number of wireless data connections used for retail applications in Europe can be expected to rise from less than 1 million to around 3 million”, M2 PressWire, (Feb. 28, 2006), 3 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 17/973,507, Jan. 26, 2024, 9 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 15/956,741, Jan. 18, 2024, 18 pgs. cited by applicant

IEEE Standard for Information technology—Telecommunication and information exchange between systems—Local and metropolitan area networks—Specific requirements, Part 15.4: Wireless Medium Access Control (MAC) and Physical Layer (PHY) Specifications for Low-Rate Wireless Personal Area Networks (LR-WPANs), IEEE, The Institute of Electrical and Electronics Engineers, Inc., 3 Park Avenue, New York, NY 10016-5997, USA, Oct. 1, 2003, 679 pgs. cited by applicant

Patel et al., Notice of Allowance, U.S. Appl. No. 17/963,170, Dec. 12, 2023, 11 pgs. cited by applicant

Slick self-service options: as staffing remains lean grocers are using advanced self-service checkouts and kiosks to augment customer service, Dialog, STIC, Report Information from Dialog, Nov. 30, 2023, 8 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 18/197,071, Jan. 26, 2024, 10 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 17/985,832, Feb. 29, 2024, 10 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 17/978,894, Jan. 17, 2024, 10 pgs. cited by applicant

Mumford, “The Components of credit card vending”, Automatic Merchandiser, Oct. 9, 2009, pp. 42-47 (Year: 2009), 7 pgs. cited by applicant

“The benefits of offline vs. online vending machines”, retrieved from

<https://web.archive.org/web/20181109215952/https://www.parlevsystems.com/2018/10/23/online-offline-infographic/>, Oct. 23, 2018, 5 pgs.) (Year: 2018). cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/973,507, May 10, 2024, 10 pgs. cited by applicant

CBM2020-00026 Petition—U.S. Pat. No. 9,659,296, 99 pgs. cited by applicant

IPR2021-00086 Petition—U.S. Pat. No. 9,659,296, 74 pgs. cited by applicant

IPR2023-01186 Institution—U.S. Pat. No. 8,856,045 B1, Entered: Jan. 19, 2024, 23 pgs. cited by applicant

IPR2023-01186 Petition—U.S. Pat. No. 8,856,045, 87 pgs. cited by applicant

IPR2023-01187 Petition—U.S. Pat. No. 10,438,208, 83 pgs. cited by applicant

IPR2023-01188 Petition—U.S. Pat. No. 10,891,608, 83 pgs., cited by applicant

IPR2023-01449 Institution—U.S. Pat. No. 11,481,772 B2, Date: Apr. 12, 2024, 27 pgs. cited by applicant

IPR2023-01449 Petition—U.S. Pat. No. 11,481,772, 96 pgs. cited by applicant

PGR2021-00077 Final Decision—U.S. Pat. No. 10,719,833 B2, Date: Oct. 26, 2022, 60 pgs. cited by applicant

PGR2021-00077 Institution—U.S. Pat. No. 10,719,833 B2, Date: Oct. 28, 2021, 53 pgs. cited by applicant

PGR2021-00077 Petition—U.S. Pat. No. 10,719,833, 102 pgs. cited by applicant

PGR2021-00084 Petition—U.S. Pat. No. 10,891,608, 123 pgs. cited by applicant

PGR2021-00093 Final Decision—U.S. Pat. No. 10,891,614 B2, 70 pgs. cited by applicant

PGR2021-00093 Institution—U.S. Pat. No. 10,891,614 B2, Entered: Dec. 17, 2021, 49 pgs. cited by applicant

PGR2021-00093 Petition—U.S. Pat. No. 10,891,614, 107 pgs. cited by applicant

PGR2022-00035 Petition—U.S. Pat. No. 11,074,580, 102 pgs. cited by applicant

PGR2023-00042 Petition—U.S. Pat. No. 11,481,772, 102 pgs. cited by applicant

PGR2023-00045 Petition—U.S. Pat. No. 11,488,174, 99 pgs. cited by applicant

PGR2023-00050 Petition—U.S. Pat. No. 11,501,296, 103 pgs. cited by applicant

Planet Laundry Magazine, “Upwardly Mobile”, Smartphones and Other Mobile Devices Are Increasingly Becoming the Way Your Customers Prefer to Find—And Interact With—Your Business, Oct. 2013, “Magazine pp. 32-33 (PDF pp. 34-35)”, 68 pgs. cited by applicant

Final Office Action, U.S. Appl. No. 18/643,965, Jul. 1, 2024, 17 pgs. cited by applicant

W. Zhang and X. L. Zhang, “Design and Implementation of Automatic Vending Machine Based on the Short Message Payment,” 2010 6th International Conference on Wireless Communications Networking and Mobile Computing (WiCOM), Chengdu, China.
<https://ieeexplore.ieee.org/document/5600192?source=1Oplus> (Year: 2010), 4 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 18/643,971, Jul. 5, 2024, 11 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 18/643,979, Jul. 18, 2024, 12 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 18/643,975, Jul. 18, 2024, 12 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/973,505, Jul. 17, 2024, 12 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/978,894, Jul. 17, 2024, 10 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/968,672, Jul. 17, 2024, 12 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 17/985,832, Aug. 1, 2024, 7 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 18/643,971, Aug. 2, 2024, 11 pgs. cited by applicant

Auto Master System and QuotePro Kiosk Announce Integrated Partnership, Global Data Point, Jun. 10, 2017, 2 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 18/643,979, Aug. 21, 2024, 12 pgs. cited by applicant

Premier Food Service Selects USA Technologies as a Strategic Partner as They Move toward Connecting 100% of Their Machines to Cashless Payment, Business Wire, New York, Sep. 18, 2017, 4 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 18/643,975, Aug. 21, 2024, 11 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 18/643,961, Aug. 29, 2024, 38 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 17/985,832, Aug. 29, 2024, 12 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 18/643,968, Sep. 18, 2024, 10 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 18/643,965, Nov. 7, 2024, 15 pgs. cited by applicant

Narongrit Waraporn et al., “Virtual Credit Cards on Mobile for M-Commerce Payment”, 2009 IEEE International Conference on e-Business Engineering, Dec. 1, 2009, 6 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 18/888,102, Nov. 20, 2024, 16 pgs. cited by applicant

Patel, Corrected Notice of Allowability, U.S. Appl. No. 18/643,979, Nov. 22, 2024, 3 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 18/888,036, Nov. 26, 2024, 10 pgs. cited by applicant

Patel, Corrected Notice of Allowability, U.S. Appl. No. 17/985,832, Nov. 27, 2024, 3 pgs. cited by applicant

Patel, Corrected Notice of Allowability, U.S. Appl. No. 18/643,975, Nov. 27, 2024, 3 pgs. cited by applicant

Patel, Corrected Notice of Allowability, U.S. Appl. No. 18/643,961, Dec. 3, 2024, 2 pgs. cited by applicant

PayRange Inc., Notice of Reasons for Rejection, Japanese Patent Application No. 2023-504380, Dec. 20, 2024, 6 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 18/610,033, Feb. 27, 2025, 9 pgs. cited by applicant

Patel, Notice of Allowance, U.S. Appl. No. 18/888,036, Mar. 19, 2025, 10 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 18/963,481, Mar. 21, 2025, 24 pgs. cited by applicant

Tao-Ku Chang et al., "A Secure Cloud-Based Payment Model for M-Commerce," 2013 42nd International Conference on Parallel Processing, Lyon, France, 2013, pp. 1082-1086, 5 pgs. doi: 10.1109/ICPP.2013.129. keywords: {Mobile communication;XML;Servers;Access control;Smart phones;Mobile Payment; Web Services;Cloud Security}. cited by applicant

Payrange, Inc., PCT/US2024/057922, International Search Report and Written Opinion, Mar. 12, 2025, 15 pgs. cited by applicant

Patel, Corrected Notice of Allowability, U.S. Appl. No. 18/643,968, Feb. 4, 2025, 3 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 19/019,278, Apr. 7, 2025, 12 pgs. cited by applicant

Patel, Corrected Notice of Allowability, U.S. Appl. No. 18/643,961, May 7, 2025, 3 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 19/077,043, May 13, 2025, 12 pgs. cited by applicant

Patel, Non-Final Office Action, U.S. Appl. No. 18/636,314, May 14, 2025, 13 pgs. cited by applicant

Patel, Final Office Action, U.S. Appl. No. 18/963,481, Jul. 14, 2025, 16 pgs. cited by applicant

A. M. Elshaer et al., "IoT-Enabled E-Prescription Management and Dispensing Machine Monitoring", Published in: 2024 14th International Conference on Electrical Engineering (ICEENG), Date Added to IEEE Xplore: Jun. 25, 2024, Cairo, Egypt, 5 pgs. cited by applicant

Primary Examiner: Nigh; James D

Attorney, Agent or Firm: Morgan, Lewis & Bockius LLP

Background/Summary

PRIORITY CLAIM AND RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/620,674 (filed Jan. 12, 2024), which is hereby incorporated by reference in its entirety. This application is also a continuation-in-part of U.S. patent application Ser. No. 18/610,033 (filed Mar. 19, 2024), which is a continuation of U.S. patent application Ser. No. 17/443,802 (filed Jul. 27, 2021), which is a continuation of U.S. patent application Ser. No. 16/934,933 (filed Jul. 21, 2020), each of which is hereby incorporated by reference in its entirety. (2) This application is related to U.S. patent application Ser. No. 18/643,965 (filed Apr. 23, 2024), Ser. No. 18/952,804 (filed Nov. 19, 2024), Ser. No. 18/888,102 (filed Sep. 17, 2024), Ser. No. 18/830,424 (filed Sep. 10, 2024), Ser. No. 18/952,778 (filed Nov. 19, 2024), Ser. No. 18/951,218 (filed Nov. 18, 2024), Ser. No. 18/197,070 (filed May 14, 2023), Ser. No. 18/197,071 (filed May 14, 2023), Ser. No. 18/643,979 (filed Apr. 23, 2024), Ser. No. 18/888,036 (filed Sep. 17, 2024), and Ser. No. 18/636,314 (filed Apr. 16, 2024), each of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present application relates to the field of tool activation and cabinet locking.

BACKGROUND

(2) Retail theft is a large and growing problem accounting for over \$100 billion in annual losses for US retailers in 2022 as stated by the National Retail Federation. Certain categories of products (e.g., articles of manufacture), such as power tools, represent a large portion of the losses due to the high value of the products combined with relatively easy approaches to resell stolen goods through

e-commerce. Left with few options, retailers lock up high-theft items behind cases or cages, but this impacts legitimate consumer shopping experiences, increases retailer costs due to added labor, and decreases overall sales due to the significant consumer inconvenience.

SUMMARY

(3) The present disclosure provides technical solutions to combat the problem of retail theft or after-purchase theft of valuable products (e.g., job-site theft of power tools).

(4) One aspect of the present disclosure describes systems, devices, and methods that leverage short-range communication (e.g., Bluetooth) technology to embed controls into a product that render the product inoperable until the product has been activated (see, e.g., FIG. 1). The activation can be persistent or temporary (e.g., time-based for rentals). A product with persistent activation can subsequently be deactivated, for example, if the product was stolen. In some implementations, after the product has been activated, the product can be assigned to an owner (e.g., bound to a user account of a mobile application that is used to activate the product), who can lock or unlock the product at the owner's discretion. The product activation process disclosed herein does not prevent legitimate resale of the product, as the owner can de-assign the product from the owner's account and re-assign the product to another user account.

(5) One aspect of the present disclosure describes systems, devices, and methods that leverage short-range communication (e.g., Bluetooth) technology to embed controls into an offline display case (e.g., display cabinet) that stocks products in a retail store. In some implementations, the controls that are embedded in the offline display case comprise controls that are integrated into a customized retail display case (see, e.g., FIG. 4) that is connected to a power supply and an electronic lock that is in a locked state (i.e., the display case is locked by default). When a user with a mobile device executing an application wants to retrieve a product from the display case, the embedded controls determine whether a user account of the application executing on the mobile device is a trusted (or authorized) account. When the embedded controls determine that the user account is a trusted account, the electronic lock changes from a locked state to an unlocked state (e.g., the retail case unlocks automatically, without user intervention) and the user may retrieve the product from the display case without having to ask a store attendant for help. In some implementations, the display case is equipped with a timer that changes the state of the lock from the unlocked state to the locked state after a predefined time period. In some implementations, the display case includes a door sensor that detects whether a door of the display case is open or close. In some implementations, the display case is configured to broadcast a status of the electronic lock and/or the cabinet door. In some implementations, the door unlocks and automatically opens and closes. In another aspect, the controls that are embedded in the offline display case comprise a display case control system (see, e.g., FIG. 5) that is installed on an existing piece of display case furniture, such as a display case that is currently used in a retail store. In some implementations, the display case is communicatively connected with a security camera system. The security system can be configured to trigger or tag the recording of the display case vicinity whenever a user is engaging with the electronic locking system of the display case. In some implementation, the display case is unlocked through a payment initiated through the user's mobile device.

(6) The subject matter described herein is particularly pointed out and distinctly claimed in the concluding portion of this specification. Objectives, features, combinations, and advantages described and implied herein will be more readily understood upon consideration of the following detailed description of the invention, taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the various described implementations, reference should be made

to the Detailed Description below, in conjunction with the following drawings. Like reference numerals refer to corresponding parts throughout the drawings.

- (2) FIG. 1 is a block diagram of a product validation and activation system in accordance with some implementations.
- (3) FIG. 2 is a schematic flow diagram of a process for activating a product in accordance with some implementations.
- (4) FIGS. 3A-3B are schematic flow diagrams of a process for activating a product in accordance with some implementations.
- (5) FIG. 4 is a block diagram of a display case security system in accordance with some implementations.
- (6) FIG. 5 is a block diagram of a display case security system including a display case control system in accordance with some implementations.
- (7) FIG. 6 is a schematic flow diagram of a process for unlocking a display case in accordance with some implementations.
- (8) FIGS. 7A-7C illustrate a flowchart diagram of method for activating a product in accordance with some implementations.
- (9) FIGS. 8A-8B illustrate a flowchart diagram of method for activating a product in accordance with some implementations.
- (10) FIGS. 9A-9C illustrate a flowchart diagram of method for unlocking an offline retail display case that stocks physical products, in accordance with some implementations.
- (11) FIG. 10 illustrates a flowchart diagram of method for unlocking an offline retail display case that stocks physical products, in accordance with some implementations.
- (12) FIG. 11 illustrates an offline retail display case in accordance with some implementations.

DETAILED DESCRIPTION

- (13) FIG. 1 is a block diagram of a product validation and activation system **100** in accordance with some implementations. The validation and activation system **100** includes a product **110**, a mobile device **130**, one or more communication networks **150**, a validation server **160**, and an activation server **180**.
- (14) In some implementations, the product **110** is a tool, a toy, an electronic device, or other product that may be susceptible to theft, including retail theft and after-purchase theft. In some implementations, the product **110** is not capable of being connected to the internet. In some implementations, the product **110** is capable of being connected to the internet but does not require this capability in the product activation processes disclosed herein.
- (15) The product **110** includes one or more processors **112**, memory **114** that is coupled to an optional controller **122**, a radio module **116**, and a power supply **120**. The memory **114** includes persistent memory **115** that stores preset authorization information. In some implementations, the controller **122** controls operation of the product **110**, and the processor **112** controls operations of the radio module **116**. In some implementations, the controller **122** sets a state (e.g., a current state) of the product. In some implementations, the controller **122** stores a state (e.g., a current state) of the product in the persistent memory **115**. In some implementations, the controller **122** and/or the processor **112** are part of the radio module **116**. In some implementations, the controller **122** and/or the processor **112** are separate controller/processing modules. In some implementations, some functions of the controller **122** and/or the processor **112** are performed by a controller/processing module embedded in the radio module **116**, and some functions of the controller **122** and/or the processor **112** are performed in a separate controller/processing module.
- (16) The persistent memory **115** retains stored information even in the absence of power being provided by the power supply **120**. The power supply **120** can be a battery pack (e.g., a swappable battery pack for a cordless power tool) or an internal battery or other internal power supply that is capable of providing power to the product without the product being connected to an external power supply. In some implementations, the product **110** includes a security unit **124** for encrypting

or decrypting messages.

(17) The radio module **116** includes short-range communication capability **118**. In some implementations, the radio module **116** includes a processor and memory. In some implementations, the radio module **116** performs the controller functions. In some implementations, the product **110** only has the radio module **116** including short-range communication capability **118** and controller **122** (e.g., in a system-on-chip (SOC) implementation). In some implementations, the radio module **116** does not use or require a separate controller. In some implementations, the radio module **116**, the processor(s) **112**, the controller **122**, the memory **114** (and persistent memory **115**), and the security unit **124** are all part of an SOC implementation (e.g., an all-in-one unit).

(18) The security unit **124** may encrypt and decrypt messages between the product **110** and the validation and activation servers **160** and **180** using a unique private key, public/private keys (i.e., asymmetric cryptography), or other encryption/decryption strategies known or yet to be discovered. For example, a unique private key may be used to securely transmit encrypted messages between the product **110** and activation server **180** (although the encrypted transmissions would most likely be routed through the mobile device **130**). The server **180** stores a private key for each product **110**, and this key is only known to the product **110** and the server **180**. No intermediary is privy to this key (especially not the mobile device **130**). When the product **110** and the server **180** communicate messages (e.g., authorization request and authorization grant messages), the security unit **124** of the product **110** encrypts the message with its private key and passes the message to the mobile device **130**. The mobile device **130** (which preferably cannot decrypt the message) passes the encrypted message to the server **180**. The security unit **186** of the server **180** is able to decrypt the message using the unique private key. The security unit **186** of the server **180** uses this same unique private key to encrypt messages to the product **110** and sends the message to the mobile device **130** to relay to the product **110**, which is able to decrypt the message using the security unit **124** and the unique private key. The product **110** may also communicate with the security unit **166** of the validation server **160** using encryption and decryption as described in the example above. The security unit **124** may be a module in memory **114** or a separate security module, configured to cause the processor **112** to implement the encryption and decryption functions described herein.

(19) In some implementations, the processor(s) **112**, the memory **114**, the radio module **116**, and the security module **124** are part of an adapter module corresponding to adapter module **100** described in any of the applications incorporated by reference in paragraphs [0001]-[0002] above (e.g., adapter module **100** described in FIGS. 1-30 and corresponding disclosure of U.S. patent application Ser. No. 18/830,424, and/or adapter module **100** described in FIGS. 1-35B and corresponding disclosure of U.S. patent application Ser. No. 18/610,033), the operations of which are not repeated here for the sake of brevity.

(20) The validation server **160** includes one or more processors **162**, memory **164**, and a communications unit **168**. The communications unit **168** of the validation server **160** includes long-range communication capability **170** (e.g., cellular technology and/or Wi-Fi mechanisms). The validation server **160** also includes a security unit **166** for encrypting and decrypting messages. The security unit **166** may be a module in memory **164** or a separate security module, configured to cause the processor **162** to implement the encryption and decryption functions described herein. The functions of the validation server **160** are described below with reference to FIGS. 2 and 3A-3B.

(21) The activation server **180** includes one or more processors **182**, memory **184**, and a communications unit **188**. The communications unit **188** of the activation server **180** includes long-range communication capability **190** (e.g., cellular technology and/or Wi-Fi mechanisms). The validation server **180** also includes a security unit **186** for encrypting and decrypting messages. The security unit **186** may be a module in memory **184** or a separate security module, configured to cause the processor **182** to implement the encryption and decryption functions described herein.

The functions of the validation server **180** are described below with reference to FIGS. 2 and 3A-3B.

(22) In some implementations, the validation server **160** and the activation server **180** are part of the same server system. In some implementations, the validation server **160** and the activation server **180** are distinct server systems. In some implementations, validation server functions are managed, configured, and/or specified by a third party that provides product validation services for OEMs (original equipment manufacturers) and retailers, and the activation server functions are managed, configured, and/or specified by one or more OEMs with respect to protected products manufactured by respective OEMs.

(23) In some implementations, the validation server **160** and/or the activation server **180** correspond to server **130** described in any of the applications incorporated by reference in paragraphs [0001]-[0002] above (e.g., server **130** described in FIGS. 1-30 and corresponding disclosure of U.S. patent application Ser. No. 18/830,424, and/or server **130** described in FIGS. 1-35B and corresponding disclosure of U.S. patent application Ser. No. 18/610,033), the operations of which are not repeated here for the sake of brevity.

(24) The mobile device **130** may be a user's personal mobile device **130**. The mobile device **130** (with a processor **132** and a mobile application **136** stored in a memory **134**) acts as a communication bridge between the product **110** and the validation and activation servers **160** and **180**. The mobile device **130** and the application **136**, however, are not “trusted” in that the communications (transmissions) they pass are encrypted. Encrypted (secured) communications are undecipherable (unencryptable, unreadable, and/or unusable) by the mobile device **130**. This keeps the communications passed between the product **110** and the servers **160** and **180** secured and safe from hacking. Mobile devices include, but are not limited to smart phones, tablet or laptop computers, personal digital assistants (PDAs), smart cards, or other technology (e.g., a hardware-software combination) known or yet to be discovered that has structure and/or capabilities similar to the mobile devices described herein. The mobile device **130** preferably has an application (e.g., the application **136**) running on it. The term “app” is used broadly to include any software program(s) capable of implementing the features described herein. The phrase “mobile device” can be assumed to include the relevant app unless specifically stated otherwise. Similarly, an “app” can be assumed to be running on an associated mobile device unless specifically stated otherwise. The mobile device **130** includes a communications unit **140**, which includes mechanisms for both long-range communications (shown as the long-range communication capability **144** such as cellular, Wi-Fi, and/or other similar communication mechanisms) for communicating with the servers **160** and **180** and short-range communications (shown as the short-range communication capability **142** such as Bluetooth and/or other similar communication mechanisms) for communicating with the product **110**.

(25) The mobile device **130** and servers **160** and **180** are communicatively coupled to each other by one or more communication networks **150**. The communication network(s) **150** are configured to convey communications (messages, signals, transmissions, and so forth). The communications include various types of information and/or instructions including, but not limited to, data, commands, bits, symbols, voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, and/or any combination thereof. The communication network(s) **150** use one or more communication protocols, such as any of Wi-Fi, Bluetooth, Bluetooth Low Energy (BLE), near-field communication (NFC), ultra-wideband (UWB), radio frequency identification (RFID), infrared wireless, induction wireless, ZigBee, Z-Wave, 6LoWPAN, Thread, 4G, 5G, and the like. Such protocols may be used to send and receive the communications using one or more transmitters, receivers, or transceivers. For example, hard-wired communications (e.g., wired serial communications) may use technology appropriate for hard-wired communications, short-range communications (e.g., Bluetooth) may use technology appropriate for close communications, and long-range communications (e.g., GSM, CDMA, Wi-Fi, wide area network (WAN), local area

network (LAN), or the like) may use technology appropriate for remote communications over a distance (e.g., over the Internet). In general, the communication network(s) **150** may include or otherwise use any wired or wireless communication technology that is known or yet to be discovered.

(26) FIG. 2 is a schematic flow diagram of a process **200** for activating a product **110** in accordance with some implementations. In accordance with some implementations of the present disclosure, products stocked in a retail store (e.g., a physical store or an online store) are in an inactivated state. An “inactivated” product is one that is non-functional (or non-operational) even though it is connected to a power supply and turned on. A consumer (e.g., associated with a mobile device **130**) purchases a product through a retail point of sales (POS) or via an online e-commerce transaction. When the consumer is ready to activate the product, the purchase must be validated (e.g., by a server system) before the product can be activated.

(27) In some implementations, the process **200** executes on a product **110**, a mobile device **130** (e.g., associated with a respective consumer and executing the application **136** thereon, or an application that is associated with a manufacturer of the product, or an application that is associated with the retail store), a validation server **160**, and an activation server **180**.

(28) The mobile device **130** launches (202) the application **136**. The mobile device **130** scans (204), using short-range communication capability **142** of the communications unit **140**, for a short-range communication device such as a Bluetooth device (e.g., radio module **116** or an adapter module) which is in the product **110**.

(29) The product **110** broadcasts (206), using short-range communication capability **118**, a unique identifier for the product **110** (e.g., serial number) along with a unique authorization (auth) request. The mobile device **130** receives the unique identifier for the product **110** and the auth request via short-range communication capability **142**. In some implementations, the broadcast may include the current state of the electronic lock (e.g., locked or unlocked), and/or the status of a door (e.g., open or closed) of a display case in which the product **110** is located. In some implementations, as an alternative to the scanning (204) and broadcasting (206) steps, a user may provide the unique ID to the application **136** by, for example, scanning a barcode or a QR code that is printed on the product **110** or manually entering the unique ID.

(30) The mobile device **130** initiates (208) (e.g., via the user or consumer) an activation process using the application **136**.

(31) The mobile device **130** receives (210) a receipt code. For example, the mobile device **130** prompts the user to scan, using the mobile device **130**, a barcode or a QR code that is printed on the purchase receipt or manually enter the receipt code.

(32) The mobile device **130** encrypts (211) the unique product ID, the unique auth code, and the receipt code with the key for the mobile device.

(33) The mobile device **130** sends (212), via the long-range communication capability **144**, the unique identifier for the product and the auth request (that the mobile device **130** received from the product **110**) along with the receipt code to the validation server **160** (or validation service). The unique identifier for the product, the auth request, and the receipt code are encrypted with the key for the mobile device **130**.

(34) The validation server **160** processes (214) the received data. In some implementations, the validation server **160** confirms, in accordance with the processing, that the receipt code is valid (e.g., the product **110** has been paid and not returned, or the receipt code has not previously been used in excess of the times permitted (e.g., one time)). In some implementations, the validation server **160** may call (communicate with) an external server/service (e.g., the retailer's server system) to verify the receipt and/or payment.

(35) The validation server **160** sends (216) a request to the activation server **180**.

(36) The activation server **180** processes (218) the request from the validation server **160**. In some implementations, the activation server creates (220) an auth grant which contains the auth request

and encrypts the message (e.g., using security unit **186**) with an encryption key unique to the product **110**.

(37) The activation server **180** associates (221) the product corresponding to the unique ID to the user account of the mobile application **136** that transmitted the auth request and receipt code in step **212**.

(38) The activation server **180** sends (222) (e.g., returns) the encrypted auth grant. In some implementations, the activation server **180** sends the encrypted auth grant directly to the mobile device **130**. In some implementations, the activation server **180** sends the encrypted auth grant to the validation server **160**, which then sends it to the mobile device **130**.

(39) The mobile device **130** establishes a connection with the product **110** and sends (224) the auth grant to the product **110**.

(40) The product **110** decrypts (226) the auth grant (e.g., using security unit **124**) with its encryption key and validates that the auth request is valid (e.g., the auth request was issued by the product **110**, the auth request was returned within a predefined amount of time, and has not been previously used). Once the validation is successful, the controller **122** changes (228) the state of the product from the inactivated state to an activated state, thus allowing the product to be used. In some implementations, activating the product (e.g., changing from the inactivated state to the activated state) includes setting by the controller **122** a flag **115** in the persistent memory **114** such that once the product **110** is activated, the activation process need not happen again.

(41) In some implementations, there is a similar flow to deactivate the product (in which case the flag **115** gets unset).

(42) In some implementations, the product activation can be set for a specific time period (e.g., 24 hours or 48 hours). The product **110** (e.g., via controller **122**) can keep track of time and unset the activation flag when the time has expired.

(43) In some implementations, the verification server **160** and activation server **180** are the same server. In some implementations the verification server **160** and activation server **180** are distinct servers. In some cases, the servers are services. In some implementations, the validation server **160** functions are managed, configured, and/or specified by a third party that provides product validation services for OEMs (original equipment manufacturers) and retailers, and the activation server **180** functions are managed, configured and/or specified by one or more OEMs with respect to protected products manufactured by respective OEMs.

(44) In some implementations, the product **110** may broadcast (e.g., transmit) additional information such as battery status, run-time and usage data, device health, and/or error codes to the mobile device **130** via the short-range communication capability **118**.

(45) In some implementations, payment for the product **110** is made using the mobile device **130** (e.g., via application **136**) and the activation process **200** is initiated when the payment is completed.

(46) In some implementations, the retail store that sells the product **110** can activate the product **110** using its own device **130**. For example, the retail store can have a POS (e.g., cash register) that executes the verification and activation process (one or more of steps **202-226**) and sends the activation (228) to the product **110** during the checkout process.

(47) In some implementations, the retail store may provide an override code instead of the receipt code.

(48) In some implementations, the retail store may use its own mobile device **130** to request activation of the product **110** without going through the verification server **160** (e.g., the mobile device **130** of the retail store bypasses verification and proceeds straight to activation).

(49) In some implementations, the same receipt code can be used multiple times (e.g., when multiple products are purchased as part of the same transaction).

(50) In some implementations, the activation process that is described with reference to FIG. 2 can be passed onto different products **110** (e.g., a first product **110-1** and a second product **110-2**). For

example, a consumer may have a tool set that includes a first tool (e.g., a drill), a second tool (e.g., a circular saw), and a third tool (e.g., a reciprocating saw), but the activation process only allows for one tool to be used at any time. In this instance, the consumer can deactivate the first tool and activate the second tool after the first tool has been deactivated. In each case, a deactivation flow and an activation flow may be followed.

(51) In some implementations, the activation message can include grants of additional features or capabilities. Using a power drill as an example product, in some instances, the drill can receive a message to upgrade it (unlock features) from a single- or two-speed drill into a variable speed drill.

(52) In some implementations, the product activation process requires identity verification from a user of the mobile device **130**.

(53) FIGS. **3A-3B** are a schematic flow diagram of a process **300** for activating a product in accordance with some implementations. In some implementations, some of the steps described in FIGS. **3A-3B** can be used in combination with some of the steps described in FIG. **2**.

(54) In some implementations, the process **300** executes on a product **110**, a mobile device **130** (e.g., associated with a respective consumer and executing the application **136** thereon, or an application that is associated with a manufacturer of the product, or an application that is associated with the retail store), a validation server **160**, and an activation server **180**.

(55) The product **110** broadcasts (302) a unique ID and a unique auth request. The mobile device **130** launches (304) application **136** and scans (306) for devices (e.g., similar to steps **202-204** in FIG. **2**). The mobile device **130** initiates (308) an activation process and obtains (310) a receipt code (e.g., similar to steps **208-210** in FIG. **2**). The method proceeds with one of two alternative process flows: Option 1 (steps **312-324**) or Option 2 (steps **326-332**).

(56) In the Option 1 process flow, the mobile device **130** sends data directly to the validation server **160**. Specifically, the mobile device **130** sends (312) the receipt code to the validation server **160** for validation, and the validation server **160** verifies (314) the receipt code. Upon validating the receipt code, the validation server **160** returns (316) a validation token to the mobile device **130**, which sends (318) a validation token to the activation server **180**. The activation server **180** sends (320) the validation token to the validation server **160**, which then verifies (322) the validation token, and sends (324) confirmation to the activation server **180** that the validation token is valid.

(57) In the Option 2 process flow, the mobile device **130** sends data directly to the activation server **180**. Specifically, the mobile device **130** sends (326) the receipt code to the activation server **180**, which sends (328) the receipt code to the validation server **160** for validation. The validation server **160** verifies (330) the receipt code and, if valid, sends (332) a validation token to the activation server **180**.

(58) After step **324** or **332** (depending on the implementation), the activation server **180** associates (334) a product ID of the product **110** to a user account of the user. The activation server **180** creates and encrypts (336) an auth grant with a unique ID, an auth request, and activation parameters, and sends (338) the auth grant to the mobile device **130**, which sends (340) the auth grant to the product **110** (similar to steps **220-224** in FIG. **2**). The product **110** decrypts (342) the auth grant, reads the parameters sent with the auth grant, and activates (344) the product in accordance with the parameters (similar to steps **226-228** in FIG. **2**).

(59) Optionally, the product **110** sends (346) confirmation that the product has been activated to the mobile device **130**, which sends (348) the confirmation to the activation server **180**. The activation server **180** then records (350) that the product activation has been successful.

(60) FIG. **4** is a block diagram of a display case security system **400** including a display case **410** configured to communicate with a mobile device **130**, and a server **460** in accordance with some implementations.

(61) The display case **410** includes one or more processors **412**, memory **414**, a radio module **416**, and a power supply **420**. In some implementations, the processor **412** controls operations of the radio module **116**. In some implementations, the processor **412** and/or the memory **414** are part of

the radio module **416**. In some implementations, the processor **412** is a separate processing module. In some implementations, some functions of the processor **412** are performed by a processing module embedded in the radio module **416**, and some functions of the processor **412** are performed in a separate processing module.

(62) The radio module **416** includes short-range communication capability **418**. In some implementations, the radio module **416** includes a processor and memory. In some implementations, the radio module **416** performs the processing functions. In some implementations, the display case **410** only has the radio module **416** including short-range communication capability **418** and processor **412** (e.g., in a system-on-chip (SOC) implementation). In some implementations, the radio module **416** does not use or require a separate processor. In some implementations, the radio module **416**, the processor(s) **412**, and the memory **414** are all part of an SOC implementation (e.g., an all-in-one unit).

(63) In some implementations, the processor(s) **412**, the memory **414**, and the radio module **416** are part of an adapter module corresponding to adapter module **100** described in any of the applications incorporated by reference in paragraphs [0001]-[0002] above (e.g., adapter module **100** described in FIGS. **1-30** and corresponding disclosure of U.S. patent application Ser. No. 18/830,424, and/or adapter module **100** described in FIGS. **1-35B** and corresponding disclosure of U.S. patent application Ser. No. 18/610,033), the operations of which are not repeated here for the sake of brevity.

(64) In some implementations, the display case is connected to a power supply **420**. For example, the power supply **420** can be an external power supply with a wired connection to the display case **410** or an internal battery or other internal power supply. In some implementations, the power supply **420** is an external power supply that is used to supply power to light the display case **410**.

(65) In some implementations, the display case includes an electronic lock **422** having a relay **424** (e.g., to engage or disengage a bolt or other locking mechanism to lock or unlock an access door of the display case **410**) and a lock state sensor **426**. The lock state sensor **426** is configured to detect a state (e.g., open or closed) of the electronic lock **422**. In some implementations, the display case **410** includes a door sensor **428** that is coupled to a door of the display case **410**. The door sensor **428** is configured to detect a state of the door (e.g., open or closed). In some implementations, the display case **410** includes a door closing mechanism **430** that is coupled to the door of the display case **410**, configured to facilitate opening and/or closing of the door. For example, in accordance with the door sensor **428** indicating an open state of the door of the display case **410**, the door closing mechanism **430** causes the door to close. For example, the door-closing mechanism can include a motor and one or more actuators.

(66) The server **460** includes one or more processors **462**, memory **464**, and a communications unit **468**. The communications unit **468** of the server **460** includes long-range communication capability **470** (e.g., cellular technology and/or Wi-Fi mechanisms). In some implementations, the server **460** includes a security unit **466** for encrypting and decrypting messages. The security unit **466** may be a module in memory **464** or a separate security module, configured to cause the processor **462** to implement the encryption and decryption functions described herein. The functions of the server **460** are described below with reference to FIG. **6**.

(67) In some implementations, the server **460** corresponds to server **130** described in any of the applications incorporated by reference in paragraphs [0001]-[0002] above (e.g., server **130** described in FIGS. **1-30** and corresponding disclosure of U.S. patent application Ser. No. 18/830,424, and/or server **130** described in FIGS. **1-35B** and corresponding disclosure of U.S. patent application Ser. No. 18/610,033), the operations of which are not repeated here for the sake of brevity.

(68) The mobile device **130** may be a user's personal mobile device **130**. The mobile device **130** (with a processor **132** and a mobile application **138** stored in a memory **134**) acts as a communication bridge between the display case **410** and the server **460**. The mobile device **130** and

the application **138**, however, are not “trusted” in that the communications (transmissions) they pass are encrypted. Encrypted (secured) communications are undecipherable (unencryptable, unreadable, and/or unusable) by the mobile device **130**. This keeps the communications passed between the display case **410** and the server **460** secured and safe from hacking. Mobile devices include, but are not limited to smart phones, tablet or laptop computers, personal digital assistants (PDAs), smart cards, or other technology (e.g., a hardware-software combination) known or yet to be discovered that has structure and/or capabilities similar to the mobile devices described herein. The mobile device **130** preferably has an application (e.g., the application **138**) running on it. The term “app” is used broadly to include any software program(s) capable of implementing the features described herein. The phrase “mobile device” can be assumed to include the relevant app unless specifically stated otherwise. Similarly, an “app” can be assumed to be running on an associated mobile device unless specifically stated otherwise. The mobile device **130** includes a communications unit **140**, which includes mechanisms for both long-range communications (shown as the long-range communication capability **144** such as cellular, Wi-Fi, and/or other similar communication mechanisms) for communicating with the server **460** and short-range communications (shown as the short-range communication capability **142** such as Bluetooth and/or other similar communication mechanisms) for communicating with the display case **410**.

(69) The mobile device **130** and server **460** are communicatively coupled to each other by one or more communication networks **150**. The communication network(s) **150** are configured to convey communications (messages, signals, transmissions, and so forth). The communications include various types of information and/or instructions including, but not limited to, data, commands, bits, symbols, voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, and/or any combination thereof. The communication network(s) **150** use one or more communication protocols, such as any of Wi-Fi, Bluetooth, Bluetooth Low Energy (BLE), near-field communication (NFC), ultra-wideband (UWB), radio frequency identification (RFID), infrared wireless, induction wireless, ZigBee, Z-Wave, 6LoWPAN, Thread, 4G, 5G, and the like. Such protocols may be used to send and receive the communications using one or more transmitters, receivers, or transceivers. For example, hard-wired communications (e.g., wired serial communications) may use technology appropriate for hard-wired communications, short-range communications (e.g., Bluetooth) may use technology appropriate for close communications, and long-range communications (e.g., GSM, CDMA, Wi-Fi, wide area network (WAN), local area network (LAN), or the like) may use technology appropriate for remote communications over a distance (e.g., over the Internet). In general, the communication network(s) **150** may include or otherwise use any wired or wireless communication technology that is known or yet to be discovered.

(70) FIG. 5 illustrates a block diagram of a display case security system **500** including a display case control system **505** configured to communicate with a display case **410** and a mobile device **130**, and a server **460** in accordance with some implementations.

(71) Systems **400** (FIG. 4) and **500** (FIG. 5) are similar, except the display case control system **505** and the power supply **520** are independent modules. In some implementations, the control system **505** is a “kit” that is installed on or inside of a display case **410** (e.g., in order to retrofit an offline display case). For example, the display case **410** can be a standard piece of furniture that is commercially available and/or already exists in a retail store. In some implementations, the control system **505** and the display case **410** are connected to an external power supply **520**. In some implementations, the lock control circuit **515** is configured to control the state of the electronic lock **422**. Specifically, rather than the processor **412** directly controlling the electronic lock **422** as in system **400** (FIG. 4), the processor **412** sends instructions to the lock control circuit **515**, which transmits those instructions to the electronic lock **422** via an adapter assembly. In some implementations, the radio module **416**, the processor(s) **412**, the memory **414**, and the lock control circuit **515** are all part of an SOC implementation (e.g., an all-in-one unit).

(72) FIG. 11 illustrates an offline retail display case **1100** in accordance with some implementations. In some implementations, the offline retail display case **1100** corresponds to display case **410** and includes internal electronics **1102** comprising the processor **412**, memory **414**, and radio module **416** as described above with reference to system **400** (e.g., implemented separately or as an SOC). In some implementations, the offline retail display case **1100** corresponds to display case **410** and includes internal electronics **1102** comprising the processor **412**, memory **414**, radio module **416**, and lock control circuit **515** as described above with reference to system **500** (e.g., implemented separately or as an SOC).

(73) The display case **1100** includes a lock **1104** corresponding to or communicatively coupled to an electronic locking mechanism (e.g., electronic lock **422** described above). In some implementations, the lock **1104** is the electronic lock **422**. In some implementations, the display case **1100** includes a door sensor (e.g., **428**) and a door closing mechanism (e.g., **430**) (not shown in the figure). In some implementations, the display case **1100** includes an indicator **1106** (e.g., an LED) that indicates the lock status of the display case (e.g., red when locked and green when unlocked). In some implementations, the display case **1100** includes an identifier **1108** (e.g., a scannable barcode or QR code) that identifies the display case **1100**, or more specifically, the electronics **1102**, to the mobile device **130** and/or the server **460**.

(74) FIG. 6 is a schematic flow diagram of a process **600** for unlocking a display case **410** in accordance with some implementations. In some implementations, the process **600** executes at a display case **410** that is displayed at a retail store or in another environment in which products or goods are displayed and provided to end users, a mobile device **130** (e.g., associated with a consumer or other end user and each executing the application **138** thereon), and a server **460** that is managed by a retailer of the retail store. The server **460** includes the components and functionalities of the validation server **160**, and/or the activation server **180**. It should be apparent to one of ordinary skill in the art that the steps of the process described with respect to the display case **410** as implemented in system **400** (FIG. 4) are also applicable to the display case **410** as implemented in system **500** (FIG. 5).

(75) The display case **410** broadcasts (**602**) a unique display case identifier and a unique authorization request (e.g., via radio module **416** using short-range communication capability **418**).

(76) In some implementations, the display case **410** broadcasts (**603**) a status of a lock (e.g., electronic lock **422**) and a status of a door of the display case **410**.

(77) The mobile device **130** executing the application **138** receives the unique display case identifier and unique auth request via short range communication capability **142**. The mobile device **130** sends (**604**), via the long-range communication capability **144**, a request to the server **460** to establish a trusted relationship between a user account of the application and the server **460**. The request includes the unique display case identifier, the unique authorization request, and a user identifier corresponding to the user account of the application.

(78) The server **460** processes (**606**) the request. In some implementations, the server **460** determines, based on the user identifier corresponding to the user account of the application, whether the user identifier belongs to a user (e.g., a customer or a potential customer) that has a trusted relationship with the retailer of the server **460**. In general, different retailers can have different parameters for establishing trusted relationships (e.g., accounts) with a customer or a potential customer. Exemplary parameters for establishing trust relationships include, and are not limited to, a customer having payment credentials with the retailer (e.g., a stored credit card or prepayment on file), a customer setting up a loyalty account with the retailer (e.g., the customer downloads an app of the retailer and registers an account), or a customer verifying their identity with the retailer (e.g., the customer sends a scanned copy of their ID to the retailer).

(79) In some implementations, when the server **460** determines that the user identifier belongs to a user that has a trusted relationship with the retailer, the server **460** creates (**607**) a trusted token. For example, the trusted token can be an encrypted auth grant that includes the unique display case

identifier and the unique authorization request). The server **460** sends (**608**) the trusted token to the mobile device **130** via long-range communication capability. The mobile device **130** sends (**610**) the trusted token and an unlock message to the display case **410** via short range communication capability **142**.

(80) The display case **410** (e.g., the radio module **410**) processes (**612**) the trusted token and the unlock message.

(81) When the display case **410** determines that the trusted token (and the unlock message) is valid, the display case **410** unlocks (**614**) the electronic lock **424**. This enables the user of the mobile device **130** to retrieve the item(s) they would like to purchase from the display case **410** and pay for the item(s) at checkout.

(82) In some implementations, the display case **410** is configured to broadcast (**616**) a state of the electronic lock **422** (e.g., whether the electronic lock is in its status (open or close) via the short-range communication capability **418** of the radio module **416**.

(83) In some implementations, the display case **410** is configured to broadcast (**618**) a status of a door of the display case **410** (e.g., open or close) that is detected by the door sensor **428** via the short-range communication capability **418** of the radio module **416**.

(84) For example, in some instances, the radio module **416** may determine, based on signals received from the door sensor **428** and the lock state sensor **426**, that the display case door is open and the electronic lock is in the locked state. In this example, the radio module **416** can send a notification via short-range communication capability **418** to a mobile device associated with a store assistant, who can then check the display case **410** and/or close the door and set the electronic lock **420** to the locked state.

(85) In some implementations, the unlocking of the electronic lock **424** triggers a timer that automatically locks (**620**) the electronic lock **424** after a predefined time period (e.g., 30 seconds, one minute, or two minutes) has elapsed.

(86) In some implementations, the display case **410** is communicatively connected with a security camera system via short-range communication capability **418**. In some implementations, the security camera system is configured to trigger recording of the display case **410** when the electronic lock is in an unlocked state.

(87) FIGS. 7A-7C illustrate a flowchart diagram of method **700** for activating a product (e.g., product **110**) (e.g., an offline product) in accordance with some implementations. The product includes (**702**) a power supply (e.g., power supply **120**), a controller (e.g., controller **122**), a persistent memory (e.g., persistent memory **115**), and a radio module (e.g., radio module **116**). In some implementations, the product has short-range communication capability (e.g., short-range communication capability **118**) via the radio module and does not have long-range communication capability. In some implementations, the product has both short- and long-range communication capabilities, but does not require the long-range communication capability for the product activation processes disclosed herein.

(88) The power supply **120** provides (**704**) power to the product without the product being connected to an external power supply. For example, the power supply can be a battery pack (e.g., a swappable battery pack for a cordless power tool) or an internal power supply or an internal battery that is capable of providing power to the product without the product being connected to an external power supply.

(89) The controller **122** is configured to control (**706**) operation of the product, including setting a state (e.g., a current state) of the product to any of a plurality of states. In some implementations, the controller is powered by the power supply. The plurality of states includes an inactivated state in which one or more functions of the product are inoperable, and an activated state in which the one or more functions of the product are operable.

(90) In one example, the product is a power tool. The functions of the power tool can include drilling cutting, shaping, grinding, and/or polishing. In another example, the product is an air fryer.

The functions of the air fryer can include broiling, roasting, dehydrating, and reheating. In some implementations, the functions of the product are also referred to as product features or product capabilities. In some implementations, a product can include optional features and/or expansion features. In some implementations, the optional features and/or expansion features can be activated in accordance with the processes disclosed herein.

(91) In accordance with some implementations disclosed herein, “activating” a product associates the product (or causes the product to be associated) with a user account of the application **136**, or a user account of an application associated with the product manufacturer, or a user account of an application associated with the retail store where the product is purchased. An inactivated product is one that is not associated with any user account.

(92) In some implementations, a product that is stocked in the retail store is in an inactivated state and can be activated (e.g., via the processes described in FIGS. 2 and 3A-3B). In some implementations, an activated product can be deactivated. A deactivated product is one that is not associated with any user account and can be (re)-activated. In some implementations, re-activating a product causes the product to be re-associated with the same usual account as that of the initial (or previous) product activation. In some implementations, re-activating a product causes the product to be re-associated with a different user account from that of the initial (or previous) product activation.

(93) In some implementations disclosed herein, an owner of an activated product can deactivate the product (i.e., dissociate the product from their user account) and transfer ownership of the product to a new owner. The new owner can then activate the product, thereby causing the product to be associated with the new owner's user account.

(94) In some implementations, the plurality of states includes a locked state and an unlocked state. A product that is in a locked state is an activated device that is locked for use, but only the authorized user can unlock it (or cause the device to change from the locked state to the unlocked state).

(95) With continued reference to FIG. 7A, the persistent memory **115** is coupled to the controller **122**. The persistent memory **115** stores the state (e.g., a current state) for the product. According to the implementations disclosed herein, the persistent memory **115** retains stored information in absence of power being provided by the power supply **120**.

(96) The radio module **116** is (712) configured to receive and transmit short-range wireless messages without use of wide area network communications. For example, the short-range wireless messages are Bluetooth messages. The short-range wireless messages are not internet messages. The wide area networking communications include the internet. In some implementations, the radio module is powered by the power supply. In some implementations, the radio module comprises a low power radio to prevent it from draining the power supply.

(97) In some implementations, the radio module **116** is (714) configured to broadcast the state of the product **110**.

(98) In some implementations, the product **110** is (716) provided in a retail environment (e.g., retail setting) (e.g., for sale) in the inactivated state.

(99) In some implementations, the product is (718) one of: a tool, a toy, or an offline electronic device. In some implementations, the product does not have any long-range communication capability (i.e., is not capable of connecting to the internet). In some implementations, the product has long-range communication capability and does not require this capability to perform any of the product activation processes disclosed herein.

(100) Referring to FIG. 7B, in some implementations, prior to receiving via the radio module the activation message and the authorization token, the controller **122** is (720) configured to transmit, to an electronic device (e.g., mobile device **130**) that is communicatively connected to the radio module via short-range communication, a unique identifier (e.g., serial number) of the product and a unique authorization request (auth request). See, e.g., FIG. 2 at step **206**.

(101) When the state of the product is (722) the inactivated state, the product receives via the radio module an activation message and an authorization token. See, e.g., FIG. 2 at steps 222 and 224.

(102) In some implementations, the product 110 receives (724) via the radio module 116 the activation message in conjunction with a purchase/receipt of the product by a consumer (e.g., buyer, purchaser, user) of the product.

(103) In some implementations, the product 110 receives (726) via the radio module 116 the activation message from an electronic device via short-range communication. The electronic device is located in proximity to (e.g., within a Bluetooth range of) the product. See, e.g., FIG. 2 at step 224.

(104) In some implementations, the activation message is (728) provided by an activation server (e.g., activation server 180) in response to an inquiry from a mobile application associated with the activation server. See, e.g., FIG. 2 at steps 216, 218, and 220.

(105) The product 110 (e.g., the controller 122) attempts (730) to validate the authorization token based on the state of the product being the inactivated state.

(106) When the validation is not successful, the controller 122 rejects (732) the activation message without changing the state of the product.

(107) When the validation is successful, the controller 122 changes (734) the state of the product to the activated state. In some implementations, the controller 122 causes the radio module 116 to transmit an activation success message. The activated product is associated with (e.g., bound to, registered to) the user account of the mobile application that transmits the activation message and the authorization token (see, e.g., FIG. 2 at step 221).

(108) In some implementations, when the validation is successful, the persistent memory 115 stores (736) a flag indicating that the product is in the activated state (e.g., so that the activation process need not happen again when the registered user uses the product).

(109) Referring to FIG. 7C, in some implementations, after changing the state of the product to the activated state, and in response to a short-range wireless message received by the radio module from the electronic device, the controller sets (738) (e.g., changes) the state of the product to: a deactivated state, a locked state, and/or an unlocked state.

(110) In some implementations, the state of the product is (740) set by the controller to the unlocked state. The controller is further configured to change the state of the product from the unlocked state to the locked state after a predefined time duration.

(111) In some implementations, the state of the product is (742) set by the controller to the unlocked state. The controller is further configured to change the state of the product from the unlocked state to the locked state after the product has been used for a predefined number of times.

(112) Even though even though the product activation process that is described in FIGS. 7A to 7C discuss binding of the product to one user account, it will be apparent to one of ordinary skill in the art that in some implementations, the same product can be associated with (e.g., bound to or registered to) many user accounts at the same time. This scenario can occur, for example, in a job site setting where the same product can be used by multiple authorized users.

(113) It should be understood that the particular order in which the operations in FIGS. 7A to 7C have been described is merely exemplary and is not intended to indicate that the described order is the only order in which the operations could be performed. One of ordinary skill in the art would recognize various ways to reorder the operations described herein. Additionally, it should be noted that details of other processes described herein with respect to other methods described herein are also applicable in an analogous manner to the method 700 described above with respect to FIGS. 7A to 7C.

(114) FIGS. 8A-8B illustrate a flowchart diagram of method 800 for activating a product (e.g., an offline product) (e.g., product 110), in accordance with some implementations. Additional details of the method 800 can be found in FIGS. 1 to 7C and the accompanying descriptions, and are not repeated for the sake of brevity.

(115) The method includes associating (802) the product with preset authorization information and storing the preset authorization information in a persistent memory (e.g., persistent memory 115) of the product.

(116) In some implementations, the product includes (804) one or more radios (e.g., radio module 116) for receiving and transmitting short-range wireless messages.

(117) The method includes distributing (806) the product for sale with a state of the product set to an inactivated state. The state of the product is selected from one or more states, including the inactivated state in which one or more functions of the product are inoperable; and an activated state in which the one or more functions of the product are operable. The state is stored in the persistent memory 115 of the product.

(118) The method includes, in conjunction with an authorized sale of the product to a consumer (e.g., a purchaser, a buyer, or a user), providing (808) to an application that executes on a first electronic device of the consumer an authorization code for the product. In some implementations, the authorization code for the product is transmitted from a validation and/or activation server (e.g., validation server 160 or activation server 180).

(119) In some implementations, the authorization code is (810) provided to the electronic device by an internet message transmitted by an authorization server (via long-range communication).

(120) The method includes, in conjunction with an indication by the application that the consumer desires to activate the product, transmitting (812) to the product by the application an activation message and the authorization code.

(121) In some implementations, the activation message and the authorization code are (814) transmitted to the product via messages transmitted by a radio (e.g., communications unit 140) of the electronic device using short-range communication.

(122) In some implementations, the activation message is (816) provided by an activation server (e.g., activation server 180) associated with the application. The activation server and the first electronic device are communicatively connected via long-range communication.

(123) Referring to FIG. 8B, the method 800 includes, at the offline product, in response to receiving the activation message and the authorization code, attempting (818) to validate the activation code based on the preset authorization information and the state of the product being the inactivated state.

(124) The method includes, when the validation is not successful, rejecting (820) the activation message without changing the state of the product, and

(125) The method includes, when the validation is successful, changing (822), by the product (e.g., automatically and without user intervention), the state of the product to the activated state. In some implementations, the method includes causing (824) the product to transmit an activation success message.

(126) In some implementations, changing the state of the product to the activated state includes associating (826) (e.g., binding, establishing a binding relationship, causing the product to be registered), by an activation server associated with the application, the product with a first user account of the application that executes on the first electronic device. For example, the activation server associates the unique identifier (e.g., serial number) of the product with the first user account.

(127) In some implementations, the first user account is (828) associated with an initial user of the product.

(128) In some implementations, the method includes, subsequent to the associating, receiving (830) by the activation server an internet message from the first electronic device. The internet message includes information of a second user account of the application that executes on a second electronic device, distinct from the first electronic device.

(129) In some implementations, the second user account is (832) associated with a subsequent user that purchases the product from the initial user.

(130) In some implementations, the method includes, in response to receiving the internet message, associating **(834)** by the activation server the product with the second user account of the application that executes on the second electronic device.

(131) FIGS. **9A-9C** illustrate a flowchart diagram of a method **900** for unlocking an offline retail display case (e.g., as implemented in system **400** and/or system **500**) that stocks physical products, in accordance with some implementations. Additional details of the method **900** can be found in FIGS. **1** to **8B** and the accompanying descriptions, and are not repeated for the sake of brevity.

(132) The retail display case includes **(902)** a power supply (e.g., power supply **420** or external power supply **520**), an electronic lock (e.g., electronic lock **422**), and a radio module (e.g., radio module **416**).

(133) The power supply provides **(804)** power to the retail display case.

(134) In some implementations, the power supply is **(906)** a wired power supply (e.g., a corded power supply that is connected to an electrical outlet).

(135) In some implementations, the power supply is **(908)** a battery.

(136) The electronic lock comprises **(910)** a state (e.g., a current state). The state is selected from one or more states including: a locked state, in which access to the physical products is prevented; and an unlocked state, in which access to the physical products is enabled.

(137) In some implementations, the electronic lock is **(912)** selected from the group consisting of: a magnetic lock, an electromagnetic lock, a Bluetooth electronic lock, and a solenoid lock, and an electromechanical lock. For example, the electronic lock can be any electronic lock that is available off-the-shelf.

(138) In some implementations, the retail display case comprises **(814)** a relay (e.g., relay **424**) (e.g., an electrically operated switch) that is connected (e.g., electrically) to the electronic lock.

(139) The radio module (e.g., a short-range transceiver) is **(916)** configured to receive and transmit short-range wireless messages without use of wide area network communications and to control the state of the electronic lock via a control signal output by the radio module. For example, the short-range wireless messages are Bluetooth messages; they are not internet messages. In some implementations, the radio module is a low power radio to prevent it from draining the power supply. In some implementations, the wide area networking communications include the internet.

(140) In some implementations, the radio module includes **(918)** a processor (e.g., processor **412**) and memory (e.g., memory **414**).

(141) In some implementations, the radio module is **(920)** configured to broadcast (e.g., periodically, over time, from time to time) short-range wireless messages specifying the state of the electronic lock. See, e.g., FIG. **6** at step **616**.

(142) In some implementations, the radio module is **(922)** configured to broadcast a unique identifier of the retail display case. See, e.g., FIG. **6** at step **602**.

(143) Referring to FIG. **9B**, in some implementations, the method **900** includes, when the state of the electronic lock is the locked state, receiving **(926)** by the radio module of the display case an unlock message and a trusted token. See, e.g., FIG. **6** at step **610**.

(144) In some implementations, the radio module receives **(928)** the unlock message from an electronic device (from an application executing on the electronic device) (via a first transceiver of the electronic device with a short-range communication capability) that is located in proximity to the retail display case.

(145) In some implementations, the radio module receives **(930)** the trusted token from an electronic device that is located in proximity to the retail display case.

(146) The radio module validates **(932)** the trusted token.

(147) When (e.g., in accordance with a determination by a processor of the radio module) the validation is not successful, the radio module rejects **(934)** the unlock message without changing the current state of the electronic lock, and

(148) When the validation is successful, the radio module changes **(936)** the state of the electronic

lock from the locked state to the unlocked state.

(149) In some implementations, the validation is successful when the trusted token identifies **(938)** a user of the electronic device as either an authorized consumer or an authorized store personnel. For example, in some implementations, the user of the electronic device is deemed (e.g., by a server of the retailer that houses the retail display case) when have a loyalty account with the retailer, or status/relationship with the retailer, or payment credentials (e.g., stored credit card or prepayment on file), or when they verify their identity (e.g., by transmitting to a server of the retailer a copy of their ID).

(150) In some implementations, the user of the electronic device is **(940)** an authorized consumer who receives the trusted token from a merchant (or a retail store) that is hosting the retail display case.

(151) In some implementations, the user of the electronic device is **(942)** an authorized consumer who receives the trusted token from an activation service of a merchant that is hosting the retail display case.

(152) With continued reference to FIG. **9C**, in some implementations, the radio module, after changing the current state of the electronic lock to the unlocked state, changes **(944)** (e.g., automatically, without user intervention) the current state of the electronic lock from the unlocked state to the locked state after a predefined time duration (e.g., X seconds, Y minutes, or a predefined time duration that is specified by the retailer).

(153) In some implementations, the radio module transmits **(946)** a short-range wireless message to a camera (e.g., a security camera) in proximity to the retail display case when (whenever) the current state of the electronic lock changes from the locked state to the unlocked state.

(154) In some implementations, the retail display case includes a door sensor (e.g., door sensor **428**) coupled to a door of the retail display case. The radio module is configured to broadcast **(948)** short-range wireless messages specifying a state of the door, wherein the state of the door comprises a door open state and a door close state. See, e.g., FIG. **6** at step **620**.

(155) FIG. **10** illustrates a flowchart diagram of a method **1000** for unlocking an offline retail display case (e.g., display case **410**) that stocks physical products, in accordance with some implementations. Additional details of the method **900** can be found in FIGS. **1** to **9C** and the accompanying descriptions, and are not repeated for the sake of brevity.

(156) The method **1000** is performed **(1002)** at a display case control system (e.g., display case control system **505**) comprising a radio module (e.g., radio module **416**) configured to receive and transmit short-range wireless messages without use of wide area network communications (e.g., using short-range communication capability **418**), a lock control circuit (e.g., lock control circuit **515**) that is electrically coupled to an electronic lock (e.g., electronic lock) of the offline display case **410**, one or more processors (e.g., processor **412**), and memory (e.g., memory **414**). The electronic lock includes a state (e.g., a current state) that is either a locked state or an unlocked state.

(157) The display case control system receives **(1004)** via the radio module an unlock message and a trusted token.

(158) The display case control system validates **(1006)** the trusted token.

(159) When the validation is not successful, the display case control system rejects **(1008)** the unlock message without changing the state of the electronic lock.

(160) When the validation is successful, the display case control system changes **(1010)** the state of the electronic lock (e.g., from the locked state to the unlocked state).

(161) Note that the various implementations described above can be combined with any other implementations described herein. The features and advantages described in the specification are not all inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been principally selected for readability and

instructional purposes and may not have been selected to delineate or circumscribe the inventive subject matter.

(162) It should be noted that relative terms are meant to help in the understanding of the technology and are not meant to limit the scope of the invention. Similarly, unless specifically stated otherwise, the terms used for labels (e.g., “first,” “second,” and “third”) are meant solely for purposes of designation and not for order or limitation. The term “short” in the phrase “short-range” (in addition to having technology specific meanings) is relative to the term “long” in the phrase “long-range.”

(163) The terms “may,” “might,” “can,” and “could” are used to indicate alternatives and optional features and only should be construed as a limitation if specifically included in the claims.

(164) It should be noted that, unless otherwise specified, the term “or” is used in its nonexclusive form (e.g., “A or B” includes A, B, A and B, or any combination thereof, but it would not have to include all of these possibilities). It should be noted that, unless otherwise specified, “and/or” is used similarly (e.g., “A and/or B” includes A, B, A and B, or any combination thereof, but it would not have to include all of these possibilities). It should be noted that, unless otherwise specified, the terms “includes” and “has” mean “comprises” (e.g., a device that includes, has, or comprises A and B contains A and B, but optionally may contain C or additional components other than A and B). It should be noted that, unless otherwise specified, the singular forms “a,” “an,” and “the” refer to one or more than one, unless the context clearly dictates otherwise.

(165) It is to be understood that the inventions, examples, and implementations described herein are not limited to particularly exemplified materials, methods, and/or structures. It is to be understood that the inventions, examples, and implementations described herein are to be considered preferred inventions, examples, and implementations whether specifically identified as such or not.

(166) The terms and expressions that have been employed in the foregoing specification are used as terms of description and not of limitation, and are not intended to exclude equivalents of the features shown and described. While the above is a complete description of selected implementations of the present invention, it is possible to practice the invention using various alternatives, modifications, adaptations, variations, and/or combinations and their equivalents. It will be appreciated by those of ordinary skill in the art that any arrangement that is calculated to achieve the same purpose may be substituted for the specific embodiment shown. It is also to be understood that the following claims are intended to cover all of the generic and specific features of the invention herein described and all statements of the scope of the invention that, as a matter of language, might be said to fall therebetween.

Claims

1. A product, comprising: a power supply that provides power to the product without the product being connected to an external power supply; a controller configured to control operation of the product, including setting a state of the product to any of a plurality of states, the plurality of states including: an inactivated state in which one or more functions of the product are inoperable; and an activated state in which the one or more functions of the product are operable; a persistent memory coupled to the controller that stores the state for the product; a radio module configured to receive and transmit short-range wireless messages without use of wide area network communications; wherein, when the state of the product is the inactivated state and the product receives via the radio module an activation message and an authorization token, the controller is configured to: attempt to validate the authorization token based on the state of the product being the inactivated state; and when the validation is not successful, reject the activation message without changing the state of the product; and when the validation is successful, change the state of the product to the activated state.

2. The product of claim 1, wherein: prior to receiving via the radio module the activation message

and the authorization token, the controller is configured to: transmit, to an electronic device that is communicatively connected to the radio module via short-range communication, a unique identifier of the product and a unique authorization request.

3. The product of claim 1, wherein the radio module is configured to broadcast the state of the product.

4. The product of claim 1, wherein the product is provided in a retail environment in the inactivated state.

5. The product of claim 1, wherein the product is one of: a tool, a toy, or an offline electronic device.

6. The product of claim 1, wherein the product receives via the radio module the activation message in conjunction with a purchase/receipt of the product by a consumer of the product.

7. The product of claim 1, wherein the product receives via the radio module the activation message from an electronic device in proximity to the product.

8. The product of claim 7, wherein the controller is further configured to: after changing the state of the product to the activated state, and in response to a short-range wireless message received by the radio module from the electronic device: set the state of the product to: a deactivated state, a locked state, and/or an unlocked state.

9. The product of claim 8, wherein: the state of the product is set by the controller to the unlocked state; and the controller is further configured to change the state of the product from the unlocked state to the locked state after a predefined time duration.

10. The product of claim 8, wherein: the state of the product is set by the controller to the unlocked state; and the controller is further configured to change the state of the product from the unlocked state to the locked state after the product has been used for a predefined number of times.

11. The product of claim 1, wherein the activation message is provided by an activation server in response to an inquiry from a mobile application associated with the activation server.

12. The product of claim 1, wherein: when the validation is successful, the persistent memory stores a flag indicating that the product is in the activated state.

13. A method, comprising: at a product that includes a power supply, a controller, a persistent memory coupled to the controller, and a radio module, wherein the product is configured to be operable in a plurality of states and a state of the product is in an inactivated state: receiving via the radio module an activation message and an authorization token; attempting to validate the authorization token based on the state of the product being the inactivated state; when the validation is not successful, rejecting the activation message without changing the state of the product; and when the validation is successful, changing the state of the product from the inactivated state to an activated state.

14. A method for activating a product, comprising: associating the product with preset authorization information and storing the preset authorization information in a persistent memory of the product; distributing the product for sale with a state set to an inactivated state, wherein the state is selected from one or more states including: the inactivated state in which one or more functions of the product are inoperable; and an activated state in which the one or more functions of the product are operable, and wherein the state is stored in the persistent memory; in conjunction with an authorized sale of the product to a consumer, providing to an application that executes on a first electronic device of the consumer an authorization code for the product; in conjunction with an indication by the application that the consumer desires to activate the product, transmitting to the product by the application an activation message and the authorization code; at the product, in response to receiving the activation message and the authorization code: attempting to validate the authorization code based on the state of the product being the inactivated state; when the validation is not successful, rejecting the activation message without changing the state of the product; and when the validation is successful, changing the state of the product to the activated state.

15. The method of claim 14, wherein the product includes one or more radios for receiving and

transmitting short-range wireless messages.

16. The method of claim 14, wherein the activation message and the authorization code are transmitted to the product via messages transmitted by a radio of the first electronic device.

17. The method of claim 14, wherein the authorization code is provided to the first electronic device by an internet message transmitted by an authorization server.

18. The method of claim 14, wherein the activation message is provided by an activation server associated with the application, wherein the activation server and the first electronic device are communicatively connected via long-range communication.

19. The method of claim 14, wherein changing the state of the product to the activated state includes: associating, by an activation server associated with the application, the product with a first user account of the application that executes on the first electronic device.

20. The method of claim 19, further comprising: subsequent to the associating, receiving by the activation server an internet message from the first electronic device, the internet message including information of a second user account of the application that executes on a second electronic device, distinct from the first electronic device; and in response to receiving the internet message, associating by the activation server the product with the second user account of the application that executes on the second electronic device.

21. The method of claim 20, wherein: the first user account is associated with an initial user of the product; and the second user account is associated with a subsequent user that purchases the product from the initial user.
